Lampricide 3-trifluoromethyl-4-nitrophenol (TFM) impacts whole-organism oxygen consumption in non-target fishes

Melina Göbel[1,*], Aiden Moore[1,*], Tabitha Hafenbrak[1], Robert Lennox[1,2], Hugo Flávio[1,‡]

[1]Department of Biology, Dalhousie University, 1355 Oxford St., Halifax, NS, Canada
[2]Ocean Tracking Network, Dalhousie University, 1355 Oxford St., Halifax NS, Canada

*These two authors contributed equally to this work.
‡Corresponding author: hflavio@dal.ca

Keywords: Comparative physiology; Respirometry; Non-target; Great Lakes; Pesticide; Oxygen consumption

Abstract

The application of the lampricide TFM (3-trifluoromethyl-4-nitrophenol) to streams infested with larval sea lamprey constitutes the backbone of the Sea Lamprey Control Program in the Laurentian Great Lakes, but native species have greatly varying sensitivities to TFM. Earlier work by D'Souza and colleagues revealed that the whole-animal oxygen consumption of larval sea lamprey responded in a dose-dependent manner to increasing TFM concentrations. We set out to determine 1) if TFM exposure leads to stepwise increases in whole-animal oxygen consumption in non-target fishes, and 2) if exposure to lethal TFM concentrations leads to increases in oxygen consumption comparable to the animals' Maximum Metabolic Rate (MMR). Using intermittent-flow respirometry, we tested three non-target fish species with known varying TFM sensitivity: young-of-the-year lake sturgeon (sensitive), juvenile rainbow trout (intermediate), and juvenile smallmouth bass (resistant). We confirmed that all three species exhibit increases in oxygen consumption during exposure to increasing concentrations of TFM, but found that the relationship between MMR and oxygen consumption prior to death during lethal TFM exposure is complex and species-dependent. While this complex relationship broadens the need for species-specific testing, it opens up interesting possibilities both for (1)

© 2026. Published by The Company of Biologists
This is an Open Access article distributed under the terms of the Creative Commons Attribution License (http://creativecommons.org/licenses/by/4.0), which permits unrestricted use, distribution and reproduction in any medium provided that the original work is properly attributed

Biology Open • Accepted manuscript

in-situ testing, e.g., through the deployment of river-side field respirometry systems to monitor real-time responses of both target and non-target species to TFM treatments, and (2) ex-situ testing, e.g., through the establishment of less-lethal assays to determine TFM sensitivity under various species × temperature × water chemistry combinations.

Summary statement

The application of lampricides is essential to control sea lamprey, but non-target species may be affected. Intermittent-flow respirometry has potential as a less-lethal testing method for sensitive species.

Introduction

Spanning more than 240,000 km^2, the contiguous Laurentian Great Lakes of North America (henceforth referred to as the Great Lakes) border both Canada and the US and represent the largest system of superficial freshwater on Earth. The Great Lakes are of immense cultural and economic importance to lakeshore communities and Indigenous peoples in the region, including the Métis, First Nations, and Tribal Nations (Pirie et al., 2025; Smith & Tibbles, 1980), and support a vast recreational and commercial fisheries industry.

The accidental introduction of sea lamprey (*Petromyzon marinus*) in the 19th century destabilized the Great Lakes ecosystem and economy (Applegate, 1951). Sea lamprey gained access to the Great Lakes from the Atlantic Ocean through a series of anthropogenic canals (Applegate, 1951; Eshenroder, 2014; Gaden et al., 2021). This jawless fish tends to target larger species of fish in the Great Lakes that are important for both commercial and recreational fisheries, threatening Great Lakes economies (Gaden et al., 2021; Smith & Tibbles, 1980). The growing severity of the sea lamprey invasion led to the formation of the Great Lakes Fishery Commission (GLFC) in 1955, with a mandate to eradicate sea lamprey from the Great Lakes. The application of the lampricide TFM (3-trifluoromethyl-4-nitrophenol) to streams infested with larval sea lamprey now constitutes the backbone of the control program, as the sedentary nature of larval sea lamprey make them an ideal life stage for targeted control (Siefkes, 2017; Sullivan et al., 2021). The application of lampricides is particularly successful both because it regularly ensures the elimination of over 90% of the larval population in a stream, and because it eliminates the sea lamprey before they reach the parasitic life stage, thereby avoiding

Biology Open • Accepted manuscript

parasitic damage to native fish populations (Applegate, 1951; Siefkes, 2017; Sullivan et al., 2021).

TFM is characterized as a protonophore for its ability to alter the mitochondrial membrane's permeability to protons and dismantle the proton motive force used to produce ATP (Birceanu et al., 2009, 2011; Kawatski et al., 1974; Viant et al., 2001). The resulting decreased intermembrane proton concentration not only lowers ATP production, but also triggers an increase in proton pumping by the electron transport chain (ETC) complexes in an attempt to restore in the proton motive force (Borowiec et al., 2022; Huerta et al., 2020; Wilkie et al., 2019). Because oxygen is the final electron acceptor of the ETC, the increased proton pumping observed during TFM exposure leads to an increased mitochondrial oxygen consumption (Birceanu et al., 2011; Borowiec et al., 2022; Huerta et al., 2020). Recently, D'Souza et al. (2025) found that this increase in cellular oxygen consumption during TFM exposure in larval sea lamprey is reflected in their whole-animal oxygen consumption and can be measured using whole-animal intermittent-flow respirometry.

Sea lamprey are particularly vulnerable to the effects of TFM because they have a weak ability to biotransform and detoxify TFM with phase II biotransformation pathways in the liver (Bussy et al., 2018; Kane et al., 1994; Lech & Statham, 1975). However, non-target Great Lakes species vary greatly in their sensitivity to TFM. For example, previous research on TFM sensitivity has identified the Centrarchidae, such as the smallmouth bass, as very TFM tolerant (Boogaard et al., 2003). Salmonids have a moderate sensitivity to TFM, with studies using rainbow trout finding they can detoxify TFM through biotransformation and excretion of the molecule in the process of glucuronidation (Kane et al., 1994; Lech & Costrini, 1972; Lech & Statham, 1975). On the other hand, the threatened lake sturgeon (*Acipenser fulvescens*) is especially sensitive to TFM when under 100 mm long, with 50% lethal concentrations (LC50) similar to those of larval sea lamprey (Boogaard et al., 2003). Further, tolerance to TFM is also dependent on environmental factors such as temperature, pH, and alkalinity (Bills et al., 2003; Hunn & Allen, 1974; McDonald & Kolar, 2007), which presents challenges towards accurately predicting dangerous TFM concentrations for non-target species.

Improving our understanding of the sensitivity of native Great Lakes species to TFM fits under a primary focus of the GLFC's sea lamprey control program (GLFC, 2025). However, traditional TFM sensitivity experiments necessitate lethal endpoints for study animals (e.g., Applegate & King Jr., 1962; Bills et al., 2003; Boogaard et al., 2003). It is not yet clear if the

Biology Open • Accepted manuscript

relationship between whole-animal oxygen consumption and TFM exposure found for larval sea lamprey by D'Souza et al. (2025) can be observed in non-target species. Additionally, D'Souza et al. (2025) postulated that the oxygen consumption of larval sea lamprey at lethal TFM concentrations (before mortality) is comparable to their maximum metabolic rate (MMR), which sets forth a question of if lethal TFM sensitivity can be derived from a species' MMR.

In this study, we aimed to determine 1) if TFM exposure leads to stepwise increases in whole-animal oxygen consumption in non-target species, and 2) if exposure to lethal TFM concentrations leads to increases in oxygen consumption comparable to the animals' MMR. We hypothesize that the increases in mitochondrial oxygen consumption induced by TFM exposure will lead to an increase in oxygen consumption at the whole-body level in non-target species just as is observed in larval sea lamprey, and that during exposure to lethal concentrations of TFM, non-target Great Lakes fish species will elevate their oxygen consumption to levels that are not significantly different from MMR. We tested these hypotheses using intermittent-flow respirometry to measure the mass specific oxygen consumption ($\dot{M}_{O_2}$) of three non-target species exposed to a range of sub-lethal and lethal concentrations of TFM: young-of-the-year lake sturgeon, juvenile rainbow trout, and juvenile smallmouth bass. These three species were selected to encompass a wide range of TFM sensitivity levels, with young-of-the-year lake sturgeon being a species of high conservation interest that is highly sensitive during early life-stages, rainbow trout being an moderately resistant species that is well studied and serves as a comparable control, and juvenile smallmouthbass being a highly resistant species that serves as an interesting comparison to more sensitive models (Boogaard et al., 2003).

Methods

Animal collection and holding

Young-of-the-year lake sturgeon (n = 95, 7-13 cm, 1.6-7.3 g) were provided by the Ontario Aquaculture Research Centre, Guelph University. Hatchery-reared juvenile rainbow trout (n = 95, 10-16 cm, 11-45 g) were kindly provided by Ocean Trout Canada Inc., located in Prince Edward Island. Wild juvenile smallmouth bass (n = 78, 5-10 cm, 2-14 g) were captured by a team from Coastal Action via electrofishing in Nova Scotia. Animals were transferred to Dalhousie University by truck, in aerated fish transport totes maintained at 15°C. Each species was housed in zebrafish racks equipped with 10 L tanks in the Aquatron Laboratory, Dalhousie University. Species were kept in separate tanks and tank density was managed depending on the size of the animals (10-20 lake sturgeon, 5-6 rainbow trout, or 3-6 smallmouth bass per

Biology Open • Accepted manuscript

tank). Tanks were supplied with artificially prepared Great Lakes water (more details provided below) on a common recirculating aquaculture system, and tank temperature ranged from 14.5°C to 15.5°C. Smallmouth bass tanks were fitted with clear plastic dividers to avoid territorial aggression. Animals were fed once daily, where lake sturgeon and smallmouth bass were fed ad lib with a mixture of bloodworms and 0.5 or 1.0 mm salmonid pellets, respectively, and rainbow trout were fed a 1-3% body-weight ration of 2.0 mm salmonid pellets. Experimental procedures were approved by the Dalhousie Animal Care Committee (Animal Use Protocol No. 25-045) and adhered to the guidelines of the Canadian Council of Animal Care (CCAC).

Water chemistry

Artificial Great Lakes water was prepared to a target alkalinity of 180 mg L^{-1} $CaCO_3$ and pH 8.5-8.6, representative of levels naturally found in rivers draining to the Great Lakes. Batches of ~800L city of Halifax dechlorinated water were carbonated using compressed CO_2 for at least 24h. $CaCO_3$ (BioShop, Canada), magnesium sulfate ($MgSO_4.7H_2O$; Bioshop, Canada), magnesium chloride ($MgCl_2.6H_2O$; Bioshop, Canada), and potassium chloride (KCl; Bioshop, Canada) were then gradually mixed in. Once all the reagents had mixed, the water was aerated for 24h to remove excess CO_2 and return the water to the target pH. Alkalinity in the fish holding system was monitored regularly using a HANNA alkalinity meter (HI775) and ranged from 65 to 182 mg L^{-1} $CaCO_3$. pH in the holding system was monitored using a benchtop pH meter (AB315 pH/mV meter, equipped with a liquid-filled pH/ATC epoxy body combination electrode, Fisherbrand, Canada) and ranged between 8.4 and 8.7. Water temperature was monitored daily in the holding tanks using a Kitchenaid temperature probe and ranged from 14.5 to 15.5°C.

Experimental setup

This study used intermittent-flow respirometry, a common method of measuring oxygen consumption in aquatic animals (e.g., Hejlesen et al., 2024; M. B. S. Svendsen et al., 2016). Four different sizes of respirometers were used in this study. These were built using clear PVC piping and acrylic caps; the small model had 4 cm inner diameter and 10 cm length of open chamber space (168.5 mL including tubing), the medium, large, and extra large models had 5.2

Biology Open • Accepted manuscript

cm inner diameter and 10, 16, and 22 cm length open space, respectively (274.5, 403.6, and 532.9 mL including tubing, respectively). The small and medium models were used for lake sturgeon and smallmouth bass, while the large and extra large models were used for rainbow trout. Respirometer volume/animal volume ratios ranged between 25 and 120 for lake sturgeon, 20-88 for smallmouth bass, and 11-35 for rainbow trout. All chambers were tested for leaks by filling them and standing them outside of water. Respirometers were covered with individual opaque sleeves during trials to minimise disturbance to the fish.

The respirometers were held in a 200 L water bath connected to two separate recirculation systems: one for clean freshwater and one for TFM water (Supplementary Material 2). Two chillers (Ecoplus, ¼ HP) maintained two aerated cold water sumps (100×45×15 cm; 75 L of water) at a temperature of 14.7°C. A custom-built temperature controller used water from these cold sumps to maintain a freshwater sump and a TFM water sump at a temperature of 15 ± 0.1°C. Two four-channel aeration pumps (FEDOUR 256PH Aquarium Air Pump) were used to thoroughly aerate the water.

Each respirometer was equipped with an oxygen probe (OXFLOW-HS; PyroScience GmBH, Aachen, Germany). One in every four respirometers was equipped with a temperature probe (TDIP15; PyroScience GmbH, Aachen, Germany) and a pH probe (PHFLOW-PK8; PyroScience GmbH) to measure in-chamber temperature and pH. All probes were connected to one of four PyroScience Firesting PRO or O_2 optical meters (FSPRO-4; PyroScience GmbH, FSO2-C4; PyroScience GmbH, respectively), allowing in-chamber oxygen concentration and temperature to be measured every second, and pH every 10 seconds. Bath temperature and pH were also recorded using a benchtop pH meter (AB315 pH/mV meter, equipped with a liquid-filled pH/ATC epoxy body combination electrode, Fisherbrand, Canada). The readings were relayed in real time to a custom-made R-based integration system, which monitored oxygen levels and temperature for unexpected changes. The integration system calculated oxygen consumption slopes and controlled the intermittent-flow phases based on R^2 thresholds and minimum acceptable oxygen levels (i.e. dynamic cycling). Each chamber was equipped with a recirculation pump (HISP-01; KOOKEE) and a flush pump (AD20P-0510A; SKESTORE). The flush pumps were controlled using a custom-built cycle controller (powered by Arduino microcontroller boards). With instruction from the cycle controller, the flush pumps alternated between phases of measurement (internal recirculation and measurement of oxygen levels) and flush (flushing of cooled, oxygenated water through the chamber). Measurement phases lasted a maximum of 240 s for rainbow trout and smallmouth bass, and 480 s for lake sturgeon. The

Biology Open • Accepted manuscript

cycle controller terminates measurement phases early if two minutes have passed and the calculated R^2 exceeds 0.95, or if in-chamber oxygen concentration reaches 180 hPa (90-85% DO depending on atmospheric pressure conditions). Flush phases for all species lasted up to 120 s or until in-chamber oxygen levels stabilized. In each trial, either 6 animals of a single species, or 12 animals of two different species were tested; always with one animal per respirometer chamber (i.e., 6 or 12 chambers were used simultaneously).

Experimental procedures

This work reports the results of two types of experimental procedures (chasing trials and TFM exposure trials) performed using three different species (lake sturgeon, rainbow trout, and smallmouth bass). Experiments on rainbow trout and smallmouth bass were only performed at the population level (i.e. the individual identity of animals used in the two sets of trials was not tracked, due to space constraints), while experiments for lake sturgeon were performed both at the population level and at the individual level, where a sub-set of the animals was identified across chasing and TFM exposure trials. Statistical tests were performed separately for the population-level analyses and for the individual-level analyses (further details provided in the data analysis sections). Where possible, animals of two species were chosen for each trial, to maximise homogeneity in the experimental design.

Exhaustive chasing trials

Pre-experiment background bacterial O_2 consumption was measured for at least five cycles, each consisting of a 240 s measurement phase and a 60s flush phase. Selected animals (n = 12) were fasted 48 h prior to the experiment. After recording the background oxygen consumption, each animal was briefly weighed in a low volume of water on a precision scale to the nearest 0.01 g. Total length was measured to the nearest mm using a viewing box equipped with a ruler. The animal was then moved to the respirometer, and an opaque cover was placed on each respirometer to minimise visual disturbance. Measurement of the standard metabolic rate (SMR) for each animal took place for ~24 h (more details on metric calculations provided in the data analyses section). After the 24 h, MMR was measured by removing each fish from their respirometer one at a time, and placing it into a 20 L bucket lined in bubble-wrap. The smallmouth bass and rainbow trout were chased for 5 min while the lake sturgeon were chased for 7 min (determined as the lengths of time required to cause a loss of equilibrium in each species). Fish were swiftly returned to their respirometers (~10 s post-exercise) to start

Biology Open • Accepted manuscript

measurement. Post-exercise $\dot{M}_{O_2}$ was then measured for 12 h.

After the 12 h recovery measurement period, animals were removed from the respirometers and placed into their holding tanks to recuperate for later reuse in the TFM trials. At least five cycles of post-experiment background O_2 consumption were then measured before ending the trial. Trials were always initiated at approximately the same time of day to minimise potential impacts of circadian rhythms on the results.

TFM exposure trials

Prior to trials, oxygen and pH probes were calibrated and background oxygen consumption levels were measured for at least five cycles. Each trial consisted of three twelve-hour periods: pre-exposure, exposure, and recovery. Similar to the chasing trials, 6 to 12 animals were fasted for approximately 48 hours before each trial, then weighed in a film of water (nearest 0.01 g) and measured to the (nearest mm), before being placed in the respirometry chambers, which were covered to minimise disturbance. Oxygen consumption measurements began immediately following placement of the animals in their respective chambers, where they were left overnight for at least 12 hours for measurements of pre-exposure SMR. To begin the exposure period, the water was dosed with a predetermined concentration of TFM. The animals were exposed to TFM for 12 hours. Temperature, alkalinity, pH and TFM concentrations were tightly monitored during the exposures (supplementary material 1). Oxygen consumption was monitored constantly during trials, and animals that appeared to be nearing mortality based on levels of oxygen consumption (i.e., displayed sharp declines in $\dot{M}_{O_2}$) were removed from their chambers and euthanized using an overdose of tricaine methanesulfonate (MS-222) buffered with sodium bicarbonate. At the end of the exposure period, the water bath was drained of TFM water, rinsed, and refilled with clean, artificial Great Lakes water. To avoid oxygen concentration within the rainbow trout respirometers from dropping to dangerous levels during the water changing process, the rainbow trout respirometers were temporarily moved to a separate bath while the main system was rinsed and refilled, and the chambers were then returned to the main water bath once the system was ready. While this transfer was essential for the wellbeing of the animals, it did cause additional stress, which is visible as a high spike in metabolic activity at the beginning of the recovery period in Figure 3. The animals were allowed to recover in clean water during a subsequent 12 hour recovery period to quantify oxygen consumption levels during their recovery. After the recovery period, the animals were removed from the chambers

Biology Open • Accepted manuscript

and euthanized using an overdose of buffered MS-222. Post-trial background oxygen consumption was then measured for at least five cycles. This study was approved by the Animal Care and Use Program at Dalhousie University (AUP 25-045) and was conducted under the guidelines of the Canadian Council on Animal Care.

Small and medium respirometers were rinsed with clean water and sprayed with 70% ethanol after every trial, while large and extra large respirometers were cleaned every 1-2 trials. At the end of every week, the whole experimental system was fully drained, scrubbed, sprayed with ethanol, and allowed to fully dry overnight.

Individual comparison of chasing and TFM exposure

Exhaustive chasing and TFM exposure procedures were conducted following the methods described in sections 2.4.1 and 2.4.2 for an individual-level comparison of 24 lake sturgeon. These lake sturgeon were tracked such that the same respirometer was used in both the chasing and TFM exposure trials, to minimise potential effects of the equipment. The animals underwent a chasing trial, then were allowed to recover for 23 days before the start of the TFM exposure trials. This 23 day interval falls within the short-term five week range shown to produce repeatable estimates of metabolic rate metrics in juvenile brown trout (Salmo trutta; Norin & Malte, 2011), while giving the animals time to recover from the chasing trials before being submitted to the TFM exposure trials.

Calculations, statistics, and data analysis

$\dot{M}_{O_2}$ calculations

$\dot{M}_{O_2}$ was calculated in R V4.5.2 (R Core Team, 2025) using the R package pyroresp v 0.1.2 (Flávio, 2026). Recorded O_2 values (hPa) were converted to μmol $O_2\,g^{-1}\,h^{-1}$ using the R package respirometry v2.0.2 (Birk, 2025). Background respiration at the time of each experimental cycle was determined by linearly modelling the change in background respiration using pre- and post-background measurements of oxygen consumption. Estimated background oxygen slopes were subtracted from the calculated oxygen consumption slope of each cycle. For the chasing trials, pre-background averaged 12.8 ± 2.0, 4.4 ± 0.7, and 12.2 ± 2.3% of SMR and post-background averaged 13.6 ± 1.3, 11.8 ± 2.4, and 28.9 ± 6.5% of SMR (mean ± SEM for lake sturgeon, rainbow trout, and smallmouth bass, respectively). For the TFM trials, pre-background averaged 26.7 ± 3.2, 3.0 ± 0.5, and 11.7 ± 2.8% of SMR and post-background

Biology Open • Accepted manuscript

averaged 20.1 ± 1.7, 3.3 ± 0.3, and 15.6 ± 1.6% of SMR (mean ± SEM for lake sturgeon, rainbow trout, and smallmouth bass, respectively). To reduce the potential impacts of background on the variables of interest, animals with background levels above 100% of their respective SMR were excluded (amounting to four smallmouth bass and one lake sturgeon in the chasing trials and five lake sturgeon in the TFM trials). Cycles with an R^2 of 0.9 or higher were used in $\dot{M}_{O_2}$ determination. Oxygen slopes were converted into $\dot{M}_{O_2}$ using the equation $\dot{M}_{O_2} = S \times V \times M^{-1}$ where S is the corrected oxygen slope (μmol O_2 g^{-1} h^{-1}), V is the volume of the respirometer corrected for the mass of the animal (assuming a 1 g:1 mL animal density), and M is the mass of the animal (g). These $\dot{M}_{O_2}$ values were then used to calculate a series of variables that are described in detail below and summarised in Table 1.

Pre-chase/pre-exposure SMR calculations

SMR was calculated as the 20th percentile of the $\dot{M}_{O_2}$ measurements recorded prior to chasing (SMR_{Chase}) or prior to TFM exposure (SMR_{TFM}) using the *quantile()* function in R (Chabot et al., 2016). The pre-chasing and pre-exposure periods lasted approximately 24 and 12 h, over which over 400 and 200 $\dot{M}_{O_2}$ measurements were collected per individual, respectively (more details provided in the results). The SMR_{Chase} of one lake sturgeon was discarded as the R^2 of most measurements failed to meet the 0.9 threshold. Similarly, the SMR_{TFM} of one smallmouth bass was discarded as the chamber was found partly open during the trial.

Separate Generalized Linear Models (GLM) with Gamma distribution and log link function were applied to each species to determine if there were any effects of day since first experiment, animal mass and condition, and background oxygen consumption on SMR_{Chase} and SMR_{TFM}. Background was included as a variable on these and all subsequent models to account for potential biases introduced by higher background levels (up to 78%), which may artificially lower SMR estimates (according to unpublished simulation estimates). Model fitting was confirmed by inspecting Q–Q and residual plots using the R package gratia v0.11.2 (Simpson, 2024). ANOVA (Type III) analysis (R package car; Fox & Weisberg, 2019) were applied to assess the significance of the tested variables. A Tukey post-hoc test was performed when significant differences were found between variables (R package emmeans; Lenth & Piaskowski, 2026).

Biology Open • Accepted manuscript

MMR calculations

MMR_F (induced by forced, exhaustive exercise) was determined as the post-chase cycle with the highest $\dot{M}_{O_2}$ measurement. Further, aerobic scope (AS) was calculated by subtracting the SMR_{Chase} of each individual from its respective MMR_F. A Gamma log-link GLM was applied to determine any effects of day since first experiment, animal mass and condition, and background oxygen consumption on MMR_F. Model fitting and variable significance as tested as stated in section 2.5.2. Additionally, MMR_S was determined as the cycle with the highest level of spontaneous activity (i.e., highest $\dot{M}_{O_2}$ measurement) prior to chasing. This allowed assessing chasing efficacy (CE) as a method to elicit maximum metabolic rate by calculating the difference between MMR_S and MMR_F, as follows:

$$CE = (MMR_F - MMR_S) / MMR_S \times 100$$

Such that CE values equal or higher than 0 indicate that chasing elicits exhaustive oxygen consumption rates (i.e., MMR_F equal or higher than MMR_S) and negative CE values indicate that chasing is a sub-optimal method to elicit exhaustive oxygen consumption rates (i.e., MMR_F lower than MMR_S). One-sample t-tests were conducted to determine if CE was significantly different from 0 for each species.

Calculation of $\dot{M}_{O_2}$ changes induced by TFM exposure

Two different metrics were used to calculate the changes in $\dot{M}_{O_2}$ induced by TFM: 80th percentile of the exposure $\dot{M}_{O_2}$ readings ($\dot{M}_{O_2TFM\ q0.8}$) using the same *quantile()* function described for the calculation of SMR, and the highest $\dot{M}_{O_2}$ value recorded during exposure ($\dot{M}_{O_2TFM\ Peak}$). Additionally, delta variants of these metrics ($\Delta\dot{M}_{O_2TFM\ q0.8}$ and $\Delta\dot{M}_{O_2TFM\ Peak}$) were calculated by subtracting the SMR_{TFM} of each individual from each respective exposure variable. A full list of all variables calculated in this study is provided in Table 1. Decisions on the type of model to apply (generalized linear vs generalized additive model) and distribution (Gaussian with identity link vs Gamma with log link) were made individually for each model depending on the nature of the response variable, the linearity of the relationship between the response variable and TFM, and the inspection of Q-Q and residual plots. In all GLMs, response variables encompass TFM concentration, time (number of days since the first trial for the species), background level (as percentage in relation to the animal's respective pre-exposure SMR), and

Biology Open • Accepted manuscript

animal mass and condition. In all generalized additive models (GAM), a smoother was applied to capture the non-linear effects of TFM concentration on $\dot{M}_{O_2}$, while time, background level, and animal mass and condition were included as a fixed effects.

Population-level comparison of MMR and TFM-induced changes in $\dot{M}_{O_2}$

$\dot{M}_{O_2 TFM\ q0.8}$ and $\dot{M}_{O_2 TFM\ Peak}$ at the highest TFM exposure were compared to MMR_F with Gamma GLMs (log-link), applied individually to each species; i.e. the model tested for differences in $\dot{M}_{O_2}$ values as a function of the variable type (factorial, 3 levels). Model fit and variable significance were confirmed as described above. Where significant differences were found, a Tukey Post Hoc test was used to determine the significance of the differences among paired groups. This analysis was repeated for the $\Delta\dot{M}_{O_2 TFM\ q0.8}$, $\Delta\dot{M}_{O_2 TFM\ Peak}$, and AS of each species.

Individual-level comparison of MMR_F and TFM-induced changes in $\dot{M}_{O_2}$ in lake sturgeon

Three lake sturgeon for which the difference in recorded mass between the chasing and TFM exposure trials was unrealistically negative (< -0.9 g) were removed from the dataset as these likely reflected an error in one of the weight measurements. Further, two lake sturgeon were excluded because their chambers displayed very high background levels (>100% of SMR). As such, for 18 individually-identified lake sturgeon, calculations of the absolute difference ($\dot{M}_{O_2 TFM\ Peak} - MMR_F$; $\dot{M}_{O_2 TFM\ q0.8} - MMR_F$) and delta difference ($\Delta\dot{M}_{O_2 TFM\ Peak} - AS$; $\Delta\dot{M}_{O_2 TFM\ q0.8} - AS$) were conducted. GLMs with Bernoulli distribution and logit-link were then applied to test if the relative differences calculated above could be used as a predictor of survival to TFM exposure. TFM concentration was included as a response variable to account for the fact some animals were exposed to 7 mg L^{-1} TFM (as opposed to 8 mg L^{-1}).

Excess post-exercise/post-exposure oxygen consumption

Excess post-exercise oxygen consumption ($EPOC_{Chase}$) and excess post-exposure oxygen consumption ($EPOC_{TFM}$) were calculated as the area between post-chase $\dot{M}_{O_2}$ readings and the animal's respective SMR (Zhang et al., 2018), until $\dot{M}_{O_2}$ reached 1.1 times SMR or 12 h

Biology Open • Accepted manuscript

had passed. Gamma GLMs (log link), were used to test if TFM concentration, time (number of days since the first trial for the species), background level (as percentage in relation to the animal's respective pre-exposure SMR), and animal mass and condition impacted $EPOC_{TFM}$. Control runs (i.e., TFM 0) were excluded from these models to improve model fit.

Half-time of oxygen consumption recovery following chasing ($t_{1/2\ chase}$) or TFM exposure ($t_{1/2\ TFM}$) were calculated following the formulas provided by Endrenyi (1998). Summarily, the $\dot{M}_{O_2}$ values up until $\dot{M}_{O_2}$ reached 1.1 times the animal's respective SMR were log-transformed and fitted with a linear regression. The slope of the linear regression (m) was then used to calculate k, which in turn allows the calculation of $t_{1/2}$, as follows:

$$k = -2.303 \times m$$

$$t_{1/2} = \ln(2) \times k^{-1}$$

Gamma GLMs (log link), were used to test if TFM concentration, time (number of days since the first trial for the species), background level (as percentage in relation to the animal's respective pre-exposure SMR), and animal mass and condition impacted $t_{1/2\ TFM}$. Control runs (i.e., TFM 0) were excluded from these models to improve model fit. Further, for lake sturgeon and smallmouth bass, a second model was run which excluded the lowest TFM concentration, to confirm the absence of a relationship between $t_{1/2\ TFM}$ and TFM concentration during exposure to higher TFM concentrations.

Results

A total of 36 trials (11 chasing trials and 25 TFM exposure trials) were conducted between 2025-11-09 and 2026-02-24 (~3.5 months). A detailed list of the trials, species tested per trial, nominal and observed TFM concentrations, and the averages and 95% range of values recorded for pH, alkalinity, and TFM are provided in supplementary material 1. Summarily, average observed TFM concentrations remained within 0.2 mg L^{-1} of the nominal TFM concentrations, with little to no variation throughout the 12 hour exposure period. Alkalinity averaged 182 mg L^{-1} $CaCO_3$ during the TFM trials, and 143 mg L^{-1} $CaCO_3$ in the chasing trials, mainly due to lower alkalinity at the initial chasing trials. The alkalinity values for 4 trials (3 TFM, 1 chasing) were lost and could not be recovered, but were known to be within expected targets as the trials were performed. pH generally averaged between 8.4 and 8.6. Further run-by-run details are provided in in supplementary material 1.

Biology Open • Accepted manuscript

A summary description of all the metrics calculated in this study is provided in Table 1. Unless stated otherwise, effect sizes are described for hypothetical average animals exposed to control conditions (i.e., 0 TFM). The statistical results of all the GLMs and GAMs conducted to explore the impacts of TFM and co-variates on the various metrics calculated are provided in Table 2 (lake sturgeon models), Table 3 (rainbow trout models), and Table 4 (smallmouth bass models), with the written results highlighting the relevant findings. The statistical results of secondary tests conducted to compare two or more calculated metrics (e.g. MMR_S vs MMR_F, or MMR_F vs $\dot{M}_{O_2 TFM\ q0.8}$ and $\dot{M}_{O_2 TFM\ Peak}$) are described directly in the text.

SMR remained relatively stable throughout the experiments

Pre-chase SMR averaged 2.3 ± 0.1 μmol O_2 g^{-1} h^{-1} for lake sturgeon (n = 45, average $\dot{M}_{O_2}$ values per animal = 403), 5.1 ± 0.1 μmol O_2 g^{-1} h^{-1} for rainbow trout (n = 18, average $\dot{M}_{O_2}$ values per animal = 412), and 2.6 ± 0.1 μmol O_2 g^{-1} h^{-1} for smallmouth bass (n = 14, average $\dot{M}_{O_2}$values per animal = 447). The recorded pre-chase SMR values for lake sturgeon lowered marginally but significantly as a function of day since the first experiment (-0.014 μmol O_2 g^{-1} h^{-1} per day; ~1% of the recorded within-day variation, Table 2). Lake sturgeon chasing trials clustered on days 0-10 (average SMR 2.7 μmol O_2 g^{-1} h^{-1}) and days 70-77 (average SMR 1.9 μmol O_2 g^{-1} h^{-1}), with the animals on the second group being larger (average 3.0 vs 4.7 g for the first and second set of trials), which could have contributed to the detected difference (although neither mass nor condition were deemed significant). None of the tested variables had a significant impact on the pre-chase SMR of rainbow trout or smallmouth bass.

Pre-exposure SMR averaged 2.4 ± 0.1 μmol O_2 g^{-1} h^{-1} for lake sturgeon (n = 66, average $\dot{M}_{O_2}$ values per animal = 238), 3.9 ± 0.1 μmol O_2 g^{-1} h^{-1} for rainbow trout (n = 77, average $\dot{M}_{O_2}$ values per animal = 214), and 2.7 ± 0.1 μmol O_2 g^{-1} h^{-1} for smallmouth bass (n = 59, average $\dot{M}_{O_2}$ values per animal = 253). Unexpectedly, the model revealed a significant but small difference in pre-exposure SMR between lake sturgeon exposure groups (0.06 μmol O_2 g^{-1} h^{-1} per mg L^{-1} TFM, Table 2). However, this represents a small portion of the within-group variation in SMR for lake sturgeon (~2.5%), so we do not expect this to have a strong impact in the interpretation of the results. Similarly to the pre-chase SMR estimates, pre-exposure SMR values for lake sturgeon were also found to have lowered marginally but significantly as a function of day since the first experiment (-0.01 μmol O_2 g^{-1} h^{-1} per day). Finally, chambers with higher background levels tended to record lower SMR values for lake sturgeon, even after

Biology Open • Accepted manuscript

background correction (-0.01 μmol O_2 g^{-1} h^{-1} per percentage point increase in background relevance; range 3-73%). The same was also the case for rainbow trout (-0.05 μmol O_2 g^{-1} h^{-1} per percentage point increase in background relevance, Table 3). Both rainbow trout and smallmouth bass showed lower SMR with larger mass (-0.04 and -0.14 μmol O_2 g^{-1} h^{-1} per gram increase, respectively) and with higher condition factor (0.20 and 0.17 μmol O_2 g^{-1} h^{-1} per 0.1 increase in condition, respectively, Table 4).

Chasing does not elevate MMR_F beyond MMR_S for lake sturgen and smallmouth bass

MMR_F averaged 8.6 ± 0.2 μmol O_2 g^{-1} h^{-1} for lake sturgeon (n = 45), 23.7 ± 0.5 μmol O_2 g^{-1} h^{-1} for rainbow trout (n = 18), and 11.5 ± 0.4 μmol O_2 g^{-1} h^{-1} for smallmouth bass (n = 14). AS averaged 6.3 ± 0.2 μmol O_2 g^{-1} h^{-1} for lake sturgeon (n = 45), 18.6 ± 0.6 μmol O_2 g^{-1} h^{-1} for rainbow trout (n = 18), and 8.9 ± 0.4 μmol O_2 g^{-1} h^{-1} for smallmouth bass (n = 14). Finally, the highest spontaneous activity prior to chasing (MMR_S) averaged 9.0 ± 0.3 μmol O_2 g^{-1} h^{-1} for lake sturgeon (n = 45), 22.0 ± 0.7 μmol O_2 g^{-1} h^{-1} for rainbow trout (n = 18), and 11.1 ± 0.7 μmol O_2 g^{-1} h^{-1} for smallmouth bass (n = 14). Progression of the $\Delta\dot{M}_{O_2}$ throughout the chasing trials for the three species are shown in Figure 1. Twenty-five lake sturgeon (56%) showed higher MMR_S than MMR_F (Figure 2), with chasing efficacy (CE) averaging -1.0 ± 2.9%. Conversely, only five rainbow trout (28%) and five smallmouth bass (36%) showed higher MMR_S than MMR_F, with CE averaging 9.6 ± 3.9% and 8.3 ± 5.9%, respectively.

Similarly to the SMR estimates, MMR_F values for lake sturgeon were also found to have lowered marginally but significantly as a function of day since the first experiment (-0.02 μmol O_2 g^{-1} h^{-1} per day, ~0.3% of the within-day variation, Table 2), but this was not seen for AS. None of the tested variables significantly affected the MMR_F or AS of rainbow trout. Finally, both the MMR_F and AS of smallmouth bass increased as a function of animal mass (0.5 to 0.7 μmol O_2 g^{-1} h^{-1} per gram increase for MMR_F, and 0.6 to 0.9 μmol O_2 g^{-1} h^{-1} per gram increase for AS).

Chasing efficacy was not significantly different from 0 for lake sturgeon (one sample two-sided t-test, t = -0.35, df = 44, p = 0.72) nor for smallmouth bass (one sample two-sided t-test, t = 1.5, df = 13, p = 0.18). However, the variation in MMR_F values was significantly lower than that of the MMR_S values for both lake sturgeon (Figure 2; two-sided F-test, F = 0.56, df = 44, p = 0.011) and smallmouth bass (two-sided F-test, F = 0.31, df = 13, p = 0.044). Finally, in rainbow

Biology Open • Accepted manuscript

trout, CE was significantly higher than 0 (one sample two-sided t-test, t = 2.46, df = 17, p = 0.024), but MMR_F and MMR_S values showed similar variation (two-sided F-test, F = 0.67, df = 17, p = 0.41).

TFM exposure consistently impacts $\dot{M}_{O_2 TFM\ q0.8}$, but not always $\dot{M}_{O_2 TFM\ Peak}$

TFM exposure led to dose-dependent increases in $\dot{M}_{O_2}$ in all three species (Figure 3). Exposure TFM concentrations ranged from 0 to 8 mg L^{-1} for lake sturgeon, and 0 to 60 mg L^{-1} for rainbow trout and smallmouth bass. Notably, twelve lake sturgeon died during exposure to 8 mg L^{-1} of TFM. Further, 16 rainbow trout died during exposure (3, 4, 3, and 6 animals at 36, 48, 54, and 60 mg L^{-1} TFM, respectively). Two smallmouth bass died during exposure (one at 48mg/L and one at 60 mg L^{-1}). The delta variants of the exposure variables (namely $\Delta\dot{M}_{O_2 TFM\ q0.8}$ and $\Delta\dot{M}_{O_2 TFM\ Peak}$) followed similar trends to their absolute equivalents ($\dot{M}_{O_2 TFM\ q0.8}$ and $\dot{M}_{O_2 TFM\ Peak}$). In the interest of brevity, those are only brought up below when they differ from the absolute variables. The full list of statistical tests from GLMs and GAMs is available in Tables 2-4.

In lake sturgeon, exposure to 8 mg L^{-1} TFM raised oxygen consumption to an average $\dot{M}_{O_2 TFM\ q0.8}$ of 10.4 ± 0.5 µmol $O_2\,g^{-1}\,h^{-1}$, and an average $\dot{M}_{O_2 TFM\ Peak}$ of 11.5 ± 0.5 µmol $O_2\,g^{-1}\,h^{-1}$ (n = 20). In rainbow trout, exposure to 60 mg L^{-1} TFM raised oxygen consumption to an average $\dot{M}_{O_2 TFM\ q0.8}$ of 14.2 ± 0.4 µmol $O_2\,g^{-1}\,h^{-1}$, and an average $\dot{M}_{O_2 TFM\ Peak}$ of 15.8 ± 0.5 µmol $O_2\,g^{-1}\,h^{-1}$ (n = 6). Finally, in smallmouth bass, exposure to 60 mg L^{-1} TFM raised oxygen consumption to an average $\dot{M}_{O_2 TFM\ q0.8}$ of 9.7 ± 1.2 µmol $O_2\,g^{-1}\,h^{-1}$, and an average $\dot{M}_{O_2 TFM\ Peak}$ of 11.2 ± 1.7 µmol $O_2\,g^{-1}\,h^{-1}$ (n = 6).

TFM concentration had a strong impact on the $\dot{M}_{O_2 TFM\ q0.8}$ and $\dot{M}_{O_2 TFM\ Peak}$ of lake sturgeon, with both variables increasing linearly with exposure to increasing TFM concentrations (1.1 and 0.7 µmol $O_2\,g^{-1}\,h^{-1}$ per mg L^{-1} TFM, respectively; Figure 4; Table 2). Experimental day had a significant negative effect on $\dot{M}_{O_2 TFM\ q0.8}$ of lake sturgeon (-0.02 µmol $O_2\,g^{-1}\,h^{-1}$ per day). However, this was not the case for $\Delta\dot{M}_{O_2 TFM\ q0.8}$.

Interestingly, while the $\dot{M}_{O_2 TFM\ q0.8}$ of rainbow trout increased with increasing TFM concentrations (0.09 to 0.17 µmol $O_2\,g^{-1}\,h^{-1}$ per mg L^{-1} TFM; Figure 4; Table 3), that was not the

Biology Open • Accepted manuscript

case for $\dot{M}_{O_2TFM\ Peak}$. This is likely the result of high, naturally occurring peaks of activity regardless of TFM concentration. None of the other tested variables impacted the absolute or delta variables in rainbow trout during the TFM exposure period.

In smallmouth bass, similarly to rainbow trout, exposure to higher TFM concentrations led to stepwise increases in $\dot{M}_{O_2TFM\ q0.8}$ (0.06 to 0.14 μmol O_2 g^{-1} h^{-1} per mg L^{-1} TFM; Figure 4; Table 4), but not $\dot{M}_{O_2TFM\ Peak}$. Larger smallmouth bass exhibited lower $\dot{M}_{O_2TFM\ q0.8}$ (-0.15 to -0.26 μmol O_2 g^{-1} h^{-1} per gram increase) and $\dot{M}_{O_2TFM\ Peak}$ (-0.44 to -1.01 μmol O_2 g^{-1} h^{-1} per gram increase), but that was not the case for $\Delta\dot{M}_{O_2TFM\ Peak}$. Finally, smallmouth bass with higher condition also exhibited lower $\dot{M}_{O_2TFM\ q0.8}$ (-0.15 to -0.29 μmol O_2 g^{-1} h^{-1} per 0.1 increase in condition) and $\dot{M}_{O_2TFM\ Peak}$ (-0.37 to -0.75 μmol O_2 g^{-1} h^{-1} per gram increase), but that was not the case for neither $\Delta\dot{M}_{O_2TFM\ q0.8}$ nor $\Delta\dot{M}_{O_2TFM\ Peak}$.

The relationship between MMR_F and TFM-induced changes in $\dot{M}_{O_2}$ is species-specific

In lake sturgeon, both $\dot{M}_{O_2TFM\ q0.8}$ and $\dot{M}_{O_2TFM\ Peak}$ during exposure to 8 mg L^{-1} TFM were significantly higher than MMR_F (by 21% and 35%; GLM, $F = 19.0$, $p < 0.001$, followed by a Tukey test, $p < 0.001$ for both comparisons; Figure 4), but $\dot{M}_{O_2TFM\ q0.8}$ and $\dot{M}_{O_2TFM\ Peak}$ were not significantly different from each other (Tukey test, $p = 0.22$). For rainbow trout, this relationship was inverted, with both $\dot{M}_{O_2TFM\ q0.8}$ and $\dot{M}_{O_2TFM\ Peak}$ during exposure to 60 mg L^{-1} TFM being significantly lower than MMR_F (by 33% and 40%; GLM, $F = 95$, $p < 0.001$, followed by a Tukey test, $p < 0.001$ for both comparisons; Figure 4). Finally, the $\dot{M}_{O_2TFM\ q0.8}$ and $\dot{M}_{O_2TFM\ Peak}$ of smallmouth bass exposed to 60 mg L^{-1} of TFM was not significantly different from their respective MMR_F (GLM, $F = 1.14$, $p = 0.34$; Figure 4). In all cases, the delta variables showed the same trends as the absolute variables.

Lake sturgeon whose $\dot{M}_{O_2TFM}$ goes severely beyond their respective MMR_F are more likely to perish during TFM exposure

The difference between MMR_F and TFM-induced changes in oxygen consumption can be used to predict the likelihood of death during exposure, with all the calculated differences being revealed as good predictors of the animals' likelihood to perish during TFM exposure (GLM, p-values presented in Figure 5 for simplicity). However, these results should be taken with caution given the small number of mortalities in the dataset.

Biology Open • Accepted manuscript

Animals that survive TFM exposure quickly return to baseline oxygen consumption levels

In lake sturgeon, average $EPOC_{TFM}$ values ranged between 6.0 and 9.4 μmol O_2 g^{-1} (for 2 and 8 mg L^{-1} TFM, respectively, Figure 6, top left), with TFM concentration significantly increasing $EPOC_{TFM}$ (Table 2). Generally, $EPOC_{TFM}$ values for lake sturgeon were within the same range as $EPOC_{Chase}$, which averaged 7.6 ± 0.6 μmol O_2 g^{-1} (n = 44). In rainbow trout, average $EPOC_{TFM}$ values ranged between 10.6 and 21.52 μmol O_2 g^{-1} (for 54 and 24 mg L^{-1} TFM, respectively, Figure 6, middle left). $EPOC_{TFM}$ displayed a non-linear trend but was not significantly related to the TFM concentration the rainbow trout had been exposed to (Table 3), and was higher than their average $EPOC_{Chase}$ of 8.9 ± 0.6 μmol O_2 g^{-1} (n = 18). Finally, in smallmouth bass, average $EPOC_{TFM}$ values ranged between 3.5 and 8.9 μmol O_2 g^{-1} (for 42 and 54 mg L^{-1} TFM, respectively, Figure 6, bottom left). Similarly to rainbow trout, the $EPOC_{TFM}$ of smallmouth bass was not significantly related to the TFM concentration they had been exposed to (Table 4), but contrary to rainbow trout, $EPOC_{TFM}$ values for smallmouth bass were lower than their average $EPOC_{Chase}$ of 9.8 ± 0.3 μmol O_2 g^{-1} (n = 18).

In lake sturgeon, average $t_{1/2\ TFM}$ values ranged between 2.1 and 5.7 hours (for 6 and 2 mg L^{-1} TFM, respectively, Figure 6, top right). $t_{1/2\ TFM}$ values for lake sturgeon were generally lower than the $t_{1/2\ Chase}$, which averaged 5.0 ± 0.5 h (n = 43). TFM concentration appeared to have a weak effect on $t_{1/2\ TFM}$, but upon closer inspection that was deemed to be a pivoting effect of generally high $t_{1/2\ TFM}$ values for the animals exposed to 2 mg L^{-1} TFM. When those animals were removed from the model, the relationship clearly disappeared from the model (Table 2). In rainbow trout, average $t_{1/2\ TFM}$ values ranged between 1.2 and 4.5 hours (for 12 and 42 mg L^{-1} TFM, respectively, Figure 6, middle right). Similarly to lake sturgeon, $t_{1/2\ TFM}$ values for rainbow trout were generally lower than the $t_{1/2\ Chase}$, which averaged 3.7 ± 0.3 h (n = 18). There was a slight non-linear trend between TFM concentration and $t_{1/2\ TFM}$ for rainbow trout, with $t_{1/2\ TFM}$ increasing as TFM increased from 20 to 42 mg L^{-1} but lowering at the highest concentrations (Table 3).

In smallmouth bass, average $t_{1/2\ TFM}$ values ranged between 2.0 and 5.7 hours (for 36 and 16 mg L^{-1} TFM, respectively, Figure 6, bottom right). As above, $t_{1/2\ TFM}$ values for smallmouth bass were generally lower than the $t_{1/2\ Chase}$, which averaged 4.3 ± 0.5 h (n = 14). Like with lake sturgeon, TFM concentration appeared to have a weak effect on $t_{1/2\ TFM}$, but that was deemed to be a pivoting effect of generally high $t_{1/2\ TFM}$ values for the animals exposed to

Biology Open • Accepted manuscript

18 mg L^{-1} TFM (the lowest TFM concentration used for smallmouth bass). Upon removal of these pivotal data points from the model, the relationship between $t_{1/2}$ and TFM clearly disappeared (Table 4).

Discussion

With this study, we aimed to determine if non-target species exhibit stepwise increases in oxygen consumption ($\dot{M}_{O_2}$) during exposure to increasing concentrations of TFM, and to determine if exercise-derived MMR was comparable to the oxygen consumption levels displayed during lethal TFM exposure in three non-target fish species with varying TFM sensitivity: lake sturgeon, rainbow trout, and smallmouth bass. Using intermittent-flow respirometry, we confirmed the presence of a stepwise relationship between TFM and oxygen consumption in all three non-target species. However, contrary to originally expected, the relationship between MMR and oxygen consumption during lethal TFM exposure was complex and species-dependent. While this complex relationship broadens the need for species-specific testing, it reveals a link between the two variables that is useful for future testing of TFM sensitivity.

TFM exposure leads to stepwise increases in $\dot{M}_{O_2}$ in non-target fishes

All three non-target fish species exhibited stepwise increases in $\dot{M}_{O_2}$ as a function of exposure to increasing TFM concentrations. As a protonophore, TFM increases the permeability of the mitochondria's inner membrane, disrupting the proton gradient that powers ATP synthesis. The weakening of the proton gradient leads to an increase in the activity of electron transport chain, which requires oxygen molecules as final receptors of electrons. Ultimately, TFM increases the amount of oxygen required to convert ADP to ATP, causing a metabolical imbalance (Birceanu et al., 2009). In the face of this insult, animals may respond by 1) biotransforming and neutralizing TFM (thus restoring the original oxygen-to-ATP cost; Kane et al., 1994), switching to anaerobic energy production pathways (therefore reducing reliance on oxygen-powered ATP production; Wilkie et al., 2007), or 3) increasing oxygen uptake (i.e., paying the added oxygen cost; D'Souza et al., 2025). As such, the overall TFM tolerance of a species/life-stage is determined by the combination of several axes of resistance, with differences in the activity and efficiency of the detoxification enzyme UDP-glucuronyltransferase (UDPGT) playing the main role (Kane et al., 1994). More tolerant species, like bluegill (a Centrarchid species closely related to smallmouth bass) and rainbow trout, have considerably

Biology Open • Accepted manuscript

higher UDPGT efficiency than sea lamprey. In comparison to sea lamprey, bluegill also exhibit significantly higher expression of detoxification genes and more responsive transcripts overall after exposure to TFM (Lawrence et al., 2022). The production and storage of these TFM detoxifying enzymes is crucial for limiting internal concentrations of TFM and minimizing its effects on the mitochondria, likely allowing smallmouth bass and rainbow trout to display an exceptional resistance to TFM compared to sensitive species like sea lamprey and lake sturgeon.

Lake sturgeon were the most TFM-sensitive of the three tested species, with sensitivity levels comparable to those of larval sea lamprey (predicted LC99.9 of 6.5-7.5 mg L-1 TFM for the water chemistry tested here; Bills et al., 2003). The high sensitivity of young lake sturgeon to TFM is well documented (Applegate & King Jr., 1962; Boogaard et al., 2003), and their $\dot{M}_{O_2}$ response to TFM exposure closely resembles that of the larval sea lamprey (D'Souza et al., 2025). Young-of-the-year lake sturgeon are in a particularly challenging position during TFM exposure because (1) they do not have as powerful a detoxification enzyme repertoire as other non-target species (Le Clair, 2014, as cited by Birceanu et al., 2021), (2) show limited capacity for increasing oxygen consumption beyond basal levels (data presented here), and (3) have not yet had the time to accumulate the energy stores required to fuel long bouts of anaerobic energy production (Ionescu et al., 2021). As such, it was expected that the lake sturgeon would show the highest TFM sensitivity levels of the three species tested here. As previously discussed, the UDPGT enzymes of highly tolerant species are more efficient than those of the lake sturgeon, and other tolerant species like bluegill exhibit significantly higher expression of detoxification genes and more responsive transcripts overall than larval sea lamprey (Kane et al., 1994; Lawrence et al., 2022). More active species, like rainbow trout and smallmouth bass, are also capable of sustaining high rates of oxygen consumption for longer periods of time than lower-activity species like the sedentary lake sturgeon. Lower availability and efficiency of TFM-detoxifying enzymes in combination with a poor aptitude to sustain high rates of oxygen consumption likely explains the high sensitivity observed in young-of-the-year lake sturgeon.

$\dot{M}_{O_2}{}^{\text{TFM q0.8}}$ is a robust metric for non-lethal TFM sensitivity tests

The 80th percentile of the oxygen consumption readings during TFM exposure ($\dot{M}_{O_2\text{TFM q0.8}}$ and $\Delta\dot{M}_{O_2\text{TFM q0.8}}$) reliably increased in a dose-dependent manner with increasing

Biology Open • Accepted manuscript

concentrations of TFM for all three species. $\dot{M}_{O_2 TFM\ q0.8}$ is resistant to sporadic cycle-to-cycle variations in $\dot{M}_{O_2}$, proving to be an effective metric for visualising TFM sensitivity. Peak oxygen consumption during TFM exposure ($\dot{M}_{O_2 TFM\ Peak}$) appears to be too heavily influenced by cycle-to-cycle variations to be a reliable metric for measuring TFM sensitivity in highly active species. Similar to the findings of D'Souza et al (2025) for larval sea lamprey, $\dot{M}_{O_2 TFM\ Peak}$ (and $\Delta\dot{M}_{O_2 TFM\ Peak}$) increased in a dose-dependent manner for lake sturgeon, but that was not the case for rainbow trout and smallmouth bass. This discrepancy may also be linked to the differing lifestyles of these species: Lake sturgeon and larval sea lamprey are both low-activity, sedentary species, while rainbow trout and smallmouth bass are high-activity, predatory species. An active lifestyle may increase the likelihood of the rainbow trout and smallmouth bass exhibiting naturally high bouts of activity during trials, which would then drive $\dot{M}_{O_2 TFM\ Peak}$ up regardless of the presence or absence of TFM. Ultimately, our results indicate $\dot{M}_{O_2 TFM\ q0.8}$ (and $\Delta\dot{M}_{O_2 TFM\ q0.8}$) are robust metrics that work across a range of species-specific activity and TFM sensitivity levels, while as $\dot{M}_{O_2 TFM\ Peak}$ (and $\Delta\dot{M}_{O_2 TFM\ Peak}$) is only a good predictor of TFM sensitivity in relatively sedentary species.

The relationship between MMR and $\dot{M}_{O_2 TFM}$ is species-specific

Lake sturgeon exhibited higher oxygen consumption values during TFM exposure than after chasing. Together with the observed poor chasing efficacy for lake sturgeon, this strongly indicates that manual chasing is not a suitable method of eliciting MMR in this species. Svendsen et al. (2014) found that exhaustive-chasing-derived MMR (MMR_F) and spontaneous MMR (MMR_S) were positively correlated in juvenile lake sturgeon, suggesting that their behaviour and physiological capacity influenced both measurements of MMR. The suitability of different methods to elicit MMR in fishes has been discussed (e.g., Killen et al., 2017; Norin & Clark, 2016). A comprehensive review by Norin and Clark (2016) found that swim-flume respirometry is a reliable method for measuring MMR in strong sustained swimmers, while exhaustive chasing is more suitable for measuring the MMR of species that do not maintain swimming for prolonged periods of time. In contrast, a review by Killen et al. (2017) found overall differences in estimated MMR between methods of forced swimming, swim flume, and exhaustive chasing to be non-critical. Lake sturgeon are considered poor swimmers across

Biology Open • Accepted manuscript

burst- and sustained-swimming categories compared to other fishes (Peake et al., 1995), likely reflecting their reliance on other physiological and morphological traits that have helped them persist through millenia (Kieffer et al., 2001). Therefore, lake sturgeon exist in a grey area between methods of estimating MMR. In this case, choosing any method and performing it well may be the most suitable solution to this dilemma (Killen et al., 2017).

Contrary to lake sturgeon, juvenile rainbow trout exhibited lower oxygen consumption values during TFM exposure than after chasing. This was a surprising result, especially given that manual chasing may not elicit true MMR in salmonids (Raby et al., 2020). Importantly, should our chasing method have underestimated MMR, then the real difference between MMR and the oxygen consumption during lethal TFM exposure is even higher than reported here. We postulate that this difference is most likely driven by the differential oxygen consumption capacity of different tissues in rainbow trout. When TFM diffuses through the body, it impacts all of the animal's tissues. However, tissue-specific sensitivity to TFM can differ, with previous research having found that depletion of anaerobic energy stores in the brain, liver, and kidneys were more significant than in the muscle (Birceanu et al., 2009, 2014; Clifford et al., 2012; Wilkie et al., 2007). Therefore, critical failure of these essential systems would lead to death without inducing MMR in rainbow trout (because MMR is driven by exhaustion of the muscle tissues). Future studies could address variation in specific-tissue oxygen consumption under TFM exposure to help elucidate the underlying mechanisms leading to death in rainbow trout before they reach MMR.

Smallmouth bass showed a strong resistance to TFM, as expected from earlier work (Boogaard et al., 2003). Unfortunately, this high resistance, compounded by a limited availability of specimens, prevented us from confidently determining lethal TFM concentrations for this species within our experimental conditions. Nevertheless, at the highest TFM concentrations tested (where limited mortality started to occur), oxygen consumption during TFM exposure approached MMR, which would be similar to what D'Souza et al. (2025) suggested to be the case for larval sea lamprey. However, fully characterizing this relationship would require further research at even higher TFM concentrations. While this would be interesting to fully understand the mechanisms that govern the relationship between whole-animal oxygen consumption and TFM exposure in resistant species, it would likely be of little application as field applications of TFM, which are usually ~80% lower than the higher concentrations tested here.

Biology Open • Accepted manuscript

The individual-level difference between MMR and oxygen consumption during TFM exposure for lake sturgeon revealed a strong potential for predicting if the animal will survive the exposure period. Interestingly, because we have found chasing to be a poor estimate of true MMR in young of the year lake sturgeon, it is possible that we could use the naturally occurring peaks in oxygen consumption as a baseline for the calculation of the difference induced by TFM exposure. This would eliminate the need for long-term tracking of individual animals in holding and also remove any confounding variables related to the progression of time. However, caution should be taken in interpreting the significance of these results due to our limited sample size for repeated measurements of MMR and TFM-induced changes, and the presence of only three mortalities within the individual-comparison dataset. Further research with a larger sample size would be required to further reinforce or reassess the significance of our results.

Non-target fishes quickly return to normal oxygen consumption after TFM exposure

All three non-target species quickly returned to SMR after being returned to freshwater, regardless of the TFM concentration they had been exposed to, which is similar to the behaviour described by D'Souza et al (2025) for larval sea lamprey. It is interesting that $t_{1/2}$ following TFM exposure was lower than $t_{1/2}$ following exhaustive chasing in all three species, highlighting the transient nature of the TFM's impacts on the animal's physiology. However, it is important to note that other long-term effects of TFM exposure have been identified. For example, long-lasting decreases in olfactory functions following TFM exposure may hinder feeding in non-target species (Sakamoto et al., 2016). Research into the potentially far-reaching effects of TFM on the physiology of non-target animals must be continued to develop a more holistic understanding of the effects of TFM on non-target species and ecosystems as a whole.

A non-lethal method for estimating TFM sensitivity

A non-lethal method of deriving TFM sensitivity could complement lethal toxicity testing and promote the sustainable use of experimental animals. Additionally, only exposing study animals to sub-lethal TFM concentrations may allow for their reuse in other experiments,

Biology Open • Accepted manuscript

helping to maximize the amount of scientific data obtained from a smaller number of study animals. Further, non-lethal testing can also be applied to the real-time monitoring of local fauna during TFM applications in the field. This would entail collecting a sample of aquatic animals from the dosing location before the TFM treatment and placing them in a field-respirometry setup, whereby their oxygen consumption can be measured using intermittent-flow respirometry. TFM application could then be monitored in real time through the oxygen consumption of these animals. Like a canary in a coal mine, a percentage of the exercise-derived MMR (or spontaneous MMR) of the sentinel animals can serve as a threshold that allows for early warning that the TFM application may be surpassing safe limits for the biological community within that habitat.

The species-dependent nature of the relationship between MMR and oxygen consumption during lethal TFM exposure limits the wide application of this method across species. Therefore, improving the robustness of this method would require testing a range of Great Lakes species that differ in physiology, life history, and ecology. It would be particularly interesting to apply this method to native Great Lakes lamprey species, which have faced large declines in abundance as a result of multiple environmental and anthropogenic stressors, including TFM applications (Marsden & Siefkes, 2019; Neave et al., 2021; Schuldt & Goold, 1980).

Using a non-lethal testing method to monitor changes in a species' TFM sensitivity will be useful in maintaining TFM treatments safe in the face of global warming. In warmer temperatures, larval sea lamprey tolerance to TFM increases as their capacity for TFM elimination is enhanced through increased TFM metabolism and excretion (Hlina et al., 2021; Mitrovic et al., 2026; Muhametsafina et al., 2019). Therefore, current sea lamprey control measures may be undermined by warmer water temperatures that last further into the TFM dosing period (September-October; Schueller et al., 2024) or increased transient elevations in temperature during TFM applications, which ultimately alter the safety TFM exposure for non-target species (Lennox et al., 2020). This is particularly relevant as sea lamprey have a very wide thermal width, with aerobic scope steadily increasing up until 28°C (H. Flávio and M.P. Wilkie, Unpublished data, as shown in Wilkie et al., 2022), a factor that likely contributed to their successful invasion of the great lakes (Wilkie, 2025). Because the thermal tolerance of native species is likely lower than that of sea lamprey (e.g. thermal optima of young-of-the-year lake sturgeon is estimated to be ~23°C, Koledin, 2024; but their critical thermal maximum may reach up to 34°C, Bugg et al., 2020) the safety margins between the dose required to kill larval sea

Biology Open • Accepted manuscript

lamprey and the dose tolerated by native species are likely to shrink in warmer conditions, and real-time monitoring will become an important tool to avoid non-target mortality events.

Conclusion

Our study has confirmed that the dose-dependent relationship between whole-animal oxygen consumption and TFM exposure is not just limited to larval sea lamprey (the target species) but is also present in non-target fish species across a wide range of TFM sensitivity levels. Further, we have unveiled species-dependent relationships between MMR and oxygen consumption during lethal TFM exposure that open interesting avenues for further research. Taken together, our results build upon the pilot work of D'Souza et al. (2025) and confirm that intermittent-flow respirometry is a viable non-lethal method to assess TFM sensitivity in non-target species. Considering that the toxicity of TFM is highly dependent on water chemistry and temperature, future research should focus on developing a field-ready, real-time monitoring system that relies on the oxygen consumption response of sentinel organisms to determine the risk of overdose and prevent non-target mortality.

Author Contributions Statement

HF conceptualized the project and methodology. MG, AM, TH, and HF conducted the experiments. HF led the data analysis, with assistance from MG and AM. MG and AM led the writing of the first manuscript draft, which was then revised by HF. All authors reviewed and contributed to the final draft.

Data Availability

The respirometry data collected for this study and the respective R analysis scripts are available as a Zenodo repository: https://doi.org/10.5281/zenodo.22956090

Biology Open • Accepted manuscript

Declaration of Competing Interests

The authors declare no competing interests.

Acknowledgements

This work was supported by a Great Lakes Fisheries Commission pilot grant awarded to HF (2025_FLA_541036) and by a Canada Foundation for Innovation John Evans Leadership Fund (JELF) awarded to RJL. We would like to thank Dr. Michael Wilkie and Marcia Chiasson for kindly providing the lake sturgeon; Aaron Craig and Ocean Trout Canada for kindly providing the rainbow trout; and Taylor Wilson, Mathieu Goldbold-Smith and the Coastal Action teams for kindly collecting and delivering the smallmouth bass used in this study. Further, we would like to thank Emily Allen, Nayla Sernowsky, Jim Eddington, Lee Jarrett, and John Batt from the Aquatron team (Dalhousie University) for their invaluable help in keeping these animals alive, mimicking Great Lakes water conditions, and ensuring the experiments succeeded.

References

Applegate, V. C. (1951). The Sea Lamprey in the Great Lakes. *The Scientific Monthly*, *72*(5), 275–281.

Applegate, V. C., & King Jr., E. L. (1962). Comparative Toxicity of 3-Trifluormethyl-4-nitrophenol (TFM) to Larval Lampreys and Eleven Species of Fishes. *Transactions of the American Fisheries Society*, *91*(4), 342–345. https://doi.org/10.1577/1548-8659(1962)91%5B342:CTOTTT%5D2.0.CO;2

Bills, T. D., Boogaard, M. A., Johnson, D. A., Brege, D. C., Scholefield, R. J., Wayne Westman, R., & Stephens, B. E. (2003). Development of a pH/Alkalinity Treatment Model for Applications of the Lampricide TFM to Streams Tributary to the Great Lakes. *Journal of Great Lakes Research, Sea Lamprey International Symposium (SLIS II)*, *29*, 510–520. https://doi.org/10.1016/S0380-1330(03)70512-7

Biology Open • Accepted manuscript

Birceanu, O., McClelland, G. B., Wang, Y. S., Brown, J. C. L., & Wilkie, M. P. (2011). The lampricide 3-trifluoromethyl-4-nitrophenol (TFM) uncouples mitochondrial oxidative phosphorylation in both sea lamprey (*Petromyzon marinus*) and TFM-tolerant rainbow trout (*Oncorhynchus mykiss*). *Comparative Biochemistry and Physiology Part C: Toxicology & Pharmacology*, *153*(3), 342–349. https://doi.org/10.1016/j.cbpc.2010.12.005

Birceanu, O., McClelland, G. B., Wang, Y. S., & Wilkie, M. P. (2009). Failure of ATP supply to match ATP demand: The mechanism of toxicity of the lampricide, 3-trifluoromethyl-4-nitrophenol (TFM), used to control sea lamprey (*Petromyzon marinus*) populations in the Great Lakes. *Aquatic Toxicology*, *94*(4), 265–274. https://doi.org/10.1016/j.aquatox.2009.07.012

Birceanu, O., Sorensen, L. A., Henry, M., McClelland, G. B., Wang, Y. S., & Wilkie, M. P. (2014). The effects of the lampricide 3-trifluoromethyl-4-nitrophenol (TFM) on fuel stores and ion balance in a non-target fish, the rainbow trout (*Oncorhynchus mykiss*). *Comparative Biochemistry and Physiology Part C: Toxicology & Pharmacology*, *160*, 30–41. https://doi.org/10.1016/j.cbpc.2013.10.002

Birceanu, O., Tessier, L. R., Huerta, B., Li, W., McDonald, A., & Wilkie, M. P. (2021). At the intersection between toxicology and physiology: What we have learned about sea lampreys and bony fish physiology from studying the mode of action of lampricides. *Journal of Great Lakes Research, Supplement on Sea Lamprey International Symposium III (SLIS III)*, *47*, S673–S689. https://doi.org/10.1016/j.jglr.2021.07.007

Birk, M. A. (2025). *respirometry: Tools for Conducting and Analyzing Respirometry Experiments*. https://doi.org/10.32614/CRAN.package.respirometry

Boogaard, M. A., Bills, T. D., & Johnson, D. A. (2003). Acute Toxicity of TFM and a TFM/Niclosamide Mixture to Selected Species of Fish, Including Lake Sturgeon (Acipenser fulvescens) and Mudpuppies (Necturus maculosus), in Laboratory and Field Exposures. *Journal of Great Lakes Research*, *29*, 529–541. https://doi.org/10.1016/S0380-1330(03)70514-0

Biology Open • Accepted manuscript

Borowiec, B. G., Birceanu, O., Wilson, J. M., McDonald, A. E., & Wilkie, M. P. (2022). Niclosamide Is a Much More Potent Toxicant of Mitochondrial Respiration than TFM in the Invasive Sea Lamprey (*Petromyzon marinus*). *Environmental Science & Technology*, *56*(8), 4970–4979. https://doi.org/10.1021/acs.est.1c07117

Bugg, W. S., Yoon, G. R., Schoen, A. N., Laluk, A., Brandt, C., Anderson, W. G., & Jeffries, K. M. (2020). Effects of acclimation temperature on the thermal physiology in two geographically distinct populations of lake sturgeon (Acipenser fulvescens). *Conservation Physiology*, *8*(1), coaa087. https://doi.org/10.1093/conphys/coaa087

Bussy, U., Chung-Davidson, Y.-W., Buchinger, T., Li, K., Smith, S. A., Jones, A. D., & Li, W. (2018). Metabolism of a sea lamprey pesticide by fish liver enzymes part A: Identification and synthesis of TFM metabolites. *Analytical and Bioanalytical Chemistry*, *410*(6), 1749–1761. https://doi.org/10.1007/s00216-017-0830-8

Chabot, D., Steffensen, J. F., & Farrell, A. P. (2016). The determination of standard metabolic rate in fishes. *Journal of Fish Biology*, *88*(1), 81–121. https://doi.org/10.1111/jfb.12845

Clifford, A. M., Henry, M., Bergstedt, R., McDonald, D. G., Smits, A. S., & Wilkie, M. P. (2012). Recovery of Larval Sea Lampreys from Short-Term Exposure to the Pesticide 3-Trifluoromethyl-4-Nitrophenol: Implications for Sea Lamprey Control in the Great Lakes. *Transactions of the American Fisheries Society*, *141*(6), 1697–1710. https://doi.org/10.1080/00028487.2012.713887

D'Souza, L., Flávio, H., & Wilkie, M. P. (2025). The lampricide 3-trifluoromethyl-4-nitrophenol (TFM) stimulates oxygen consumption by larval sea lamprey in a dose-dependent manner. *Journal of Great Lakes Research*, *51*(2), 102536. https://doi.org/10.1016/j.jglr.2025.102536

Eshenroder, R. L. (2014). The Role of the Champlain Canal and Erie Canal as Putative Corridors for Colonization of Lake Champlain and Lake Ontario by Sea Lampreys. *Transactions of the American Fisheries Society*, *143*(3), 634–649. https://doi.org/10.1080/00028487.2013.879818

Flávio, H. (2026). *pyroresp: Analyse respirometry data captured by firesting loggers* (Version 0.1.2) [Computer software]. https://github.com/hugomflavio/pyroresp

Biology Open • Accepted manuscript

Fox, J., & Weisberg, S. (2019). *An R Companion to Applied Regression* (Third). Sage. https://www.john-fox.ca/Companion/

Gaden, M., Brant, C., Stedman, R. C., Cooke, S. J., Young, N., Lauber, T. B., Nguyen, V. M., Connelly, N. A., & Knuth, B. (2021). Shifting baselines and social license to operate: Challenges in communicating sea lamprey control. *Journal of Great Lakes Research, Supplement on Sea Lamprey International Symposium III (SLIS III)*, *47*, S800–S808. https://doi.org/10.1016/j.jglr.2021.01.016

GLFC. (2025). *Sea Lamprey Control Board Research Priorities (updated October 2025)* (p. 10). https://www.glfc.org/pubs/pdfs/research/SLCB_research_priorities.pdf

Hejlesen, R., Scheffler, F. B., Byrge, C. G., Kjær-Sørensen, K., Oxvig, C., Fago, A., & Malte, H. (2024). Assessing metabolic rates in zebrafish using a 3D-printed intermittent-flow respirometer and swim tunnel system. *Biology Open*, *13*(6), bio060375. https://doi.org/10.1242/bio.060375

Hlina, B. L., Birceanu, O., Robinson, C. S., Dhiyebi, H., & Wilkie, M. P. (2021). The relationship between thermal physiology and lampricide sensitivity in larval sea lamprey (*Petromyzon marinus*). *Journal of Great Lakes Research, Supplement on Sea Lamprey International Symposium III (SLIS III)*, *47*, S272–S284. https://doi.org/10.1016/j.jglr.2021.10.002

Huerta, B., Chung-Davidson, Y.-W., Bussy, U., Zhang, Y., Bazil, J. N., & Li, W. (2020). Sea lamprey cardiac mitochondrial bioenergetics after exposure to TFM and its metabolites. *Aquatic Toxicology*, *219*, 105380. https://doi.org/10.1016/j.aquatox.2019.105380

Hunn, J. B., & Allen, J. L. (1974). Movement of Drugs Across the Gills of Fishes. *Annual Review of Pharmacology and Toxicology*, *14*(Volume 14, 1974), 47–54. https://doi.org/10.1146/annurev.pa.14.040174.000403

Ionescu, R. A., Hepditch, S. L. J., & Wilkie, M. P. (2021). The lampricide 3-trifluoromethyl-4-nitrophenol causes temporary metabolic disturbances in juvenile lake sturgeon (Acipenser fulvescens): Implications for sea lamprey control and fish conservation. *Conservation Physiology*, *9*(1), coab069. https://doi.org/10.1093/conphys/coab069

Biology Open • Accepted manuscript

Kane, A. S., Kahng, M. W., Reimschuessel, R., Nhamburo, P. T., & Lipsky, M. M. (1994). UDP-Glucuronyltransferase Kinetics for 3-Trifluoromethyl-4-nitrophenol (TFM) in Fish. *Transactions of the American Fisheries Society*, *123*(2), 217–222. https://doi.org/10.1577/1548-8659(1994)123%3C0217:UGKFTN%3E2.3.CO;2

Kawatski, J. A., Dawson, V. K., & Reuvers, M. L. (1974). Effect of TFM and Bayer 73 on in Vivo Oxygen Consumption of the Aquatic Midge *Chironomus Tentans*. *Transactions of the American Fisheries Society*, *103*(3), 551–556. https://doi.org/10.1577/1548-8659(1974)103%3C551:EOTABO%3E2.0.CO;2

Kieffer, J. D., Wakefield, A. M., & Litvak, M. K. (2001). Physiological response to exercise in sturgeon. *The Journal of Experimental Biology*, *204*, 4281–4289.

Killen, S. S., Norin, T., & Halsey, L. G. (2017). Do method and species lifestyle affect measures of maximum metabolic rate in fishes? *Journal of Fish Biology*, *90*(3), 1037–1046. https://doi.org/10.1111/jfb.13195

Koledin, M. (2024). *The effects of warmer temperatures on the physiology of invasive sea lamprey and imperiled populations of lake sturgeon in the Laurentian Great Lakes* [MS.c., Wilfrid Laurier University]. https://scholars.wlu.ca/etd/2634/

Lawrence, M. J., Grayson, P., Jeffrey, J. D., Docker, M. F., Garroway, C. J., Wilson, J. M., Manzon, R. G., Wilkie, M. P., & Jeffries, K. M. (2022). Variation in the Transcriptome Response and Detoxification Gene Diversity Drives Pesticide Tolerance in Fishes. *Environmental Science & Technology*, *56*(17), 12137–12147. https://doi.org/10.1021/acs. est.2c00821

Le Clair, M. W. (2014). *Distribution and elimination of 3-Trifluoromethyl-4-Nitrophenol (TFM) by sea lamprey (*Petromyzon marinus*) and non-target, rainbow trout (*Oncorhynchus mykiss*) and lake sturgeon (<i>Acipenser fulvescens<i>)* [M.Sc. thesis]. Wilfrid Laurier University.

Lech, J. J., & Costrini, N. V. (1972). In vitro and in vivo metabolism of 3-trifluoromethyl-4-nitrophenol (TFM) in rainbow trout. *Comparative and General Pharmacology*, *3*(10), 160–166. https://doi.org/10.1016/0010-4035(72)90021-3

Biology Open • Accepted manuscript

Lech, J. J., & Statham, C. N. (1975). Role of glucuronide formation in the selective toxicity of 3-trifluoromethyl-4-nitrophenol (TFM) for the sea lamprey: Comparative aspects of TFM uptake and conjugation in sea lamprey and rainbow trout. *Toxicology and Applied Pharmacology*, *31*(1), 150–158. https://doi.org/10.1016/0041-008X(75)90063-0

Lennox, R. J., Bravener, G. A., Lin, H.-Y., Madenjian, C. P., Muir, A. M., Remucal, C. K., Robinson, K. F., Rous, A. M., Siefkes, M. J., Wilkie, M. P., Zielinski, D. P., & Cooke, S. J. (2020). Potential changes to the biology and challenges to the management of invasive sea lamprey Petromyzon marinus in the Laurentian Great Lakes due to climate change. *Global Change Biology*, *26*(3), 1118–1137. https://doi.org/10.1111/gcb.14957

Lenth, R. V., & Piaskowski, J. (2026). *emmeans: Estimated Marginal Means, aka Least-Squares Means*. https://doi.org/10.32614/CRAN.package.emmeans

Marsden, J. E., & Siefkes, M. J. (2019). Control of Invasive Sea Lamprey in the Great Lakes, Lake Champlain, and Finger Lakes of New York. In M. F. Docker (Ed.), *Lampreys: Biology, Conservation and Control: Volume 2* (pp. 411–479). Springer Netherlands. https://doi.org/10.1007/978-94-024-1684-8_5

McDonald, D. G., & Kolar, C. S. (2007). Research to Guide the Use of Lampricides for Controlling Sea Lamprey. *Journal of Great Lakes Research*, *33*(sp2), 20–34. https://doi.org/10.3394/0380-1330(2007)33%5B20:RTGTUO%5D2.0.CO;2

Mitrovic, D., Bragg, L. M., Flávio, H., Servos, M. R., & Wilkie, M. P. (2026). The mechanism of increased tolerance to the pesticide 3-trifluoromethyl-4-nitrophenol (TFM) in invasive sea lamprey (Petromyzon marinus) at warmer water temperatures. *Environmental Toxicology and Chemistry*, *45*(8), 2193–2208. https://doi.org/10.1093/etojnl/vgag040

Muhametsafina, A., Birceanu, O., Hlina, B. L., Tessier, L. R., & Wilkie, M. P. (2019). Warmer waters increase the larval sea lamprey's (*Petromyzon marinus*) tolerance to the lampricide 3-trifluoromethyl-4-nitrophenol (TFM). *Journal of Great Lakes Research*, *45*(5), 921–933. https://doi.org/10.1016/j.jglr.2019.07.011

Biology Open • Accepted manuscript

Neave, F. B., Booth, R. M. W., Philipps, R. R., Keffer, D. A., Bravener, G. A., & Coombs, N. (2021). Changes in native lamprey populations in the Great Lakes since the onset of sea lamprey (*Petromyzon marinus*) control. *Journal of Great Lakes Research, Supplement on Sea Lamprey International Symposium III (SLIS III)*, *47*, S378–S387. https://doi.org/10.1016/j.jglr.2020.10.005

Norin, T., & Clark, T. D. (2016). Measurement and relevance of maximum metabolic rate in fishes. *Journal of Fish Biology*, *88*(1), 122–151. https://doi.org/10.1111/jfb.12796

Norin, T., & Malte, H. (2011). Repeatability of standard metabolic rate, active metabolic rate and aerobic scope in young brown trout during a period of moderate food availability. *Journal of Experimental Biology*, *214*(10), 1668–1675. https://doi.org/10.1242/jeb.054205

Peake, S. J., Beamish, F. W. H., McKinley, R. S., Katopodis, C., & Scruton, D. A. (1995). *Swimming Performance of Lake Sturgeon,* Acipenser fulvescens (2063, p. iv + 26 pp). Can. Tech. Rep. Fish. Aquat. Sci.

Pirie, E., Serville-Tertullien, M., Buell, M.-C., Wall, B. M., & Furgal, C. (2025). Indigenous Peoples and environmental research and monitoring within the Laurentian Great Lakes Basin: A systematic map. *Ecological Solutions and Evidence*, *6*(3), e70102. https://doi.org/10.1002/2688-8319.70102

R Core Team. (2025). *R: A Language and Environment for Statistical Computing*. R Foundation for Statistical Computing. https://www.R-project.org/

Raby, G. D., Doherty, C. L. J., Mokdad, A., Pitcher, T. E., & Fisk, A. T. (2020). Post-exercise respirometry underestimates maximum metabolic rate in juvenile salmon. *Conservation Physiology*, *8*(1). https://doi.org/10.1093/conphys/coaa063

Sakamoto, K., Dew, W. A., Hecnar, S. J., & Pyle, G. G. (2016). Effects of Lampricide on Olfaction and Behavior in Young-of-the-Year Lake Sturgeon (*Acipenser fulvescens*). *Environmental Science & Technology*, *50*(7), 3462–3468. https://doi.org/10.1021/acs.est.6b01051

Biology Open • Accepted manuscript

Schueller, J. R., Boogaard, M. A., Kirkeeng, C. A., Schloesser, N. A., Wolfe, S. L., Lettenberger, A. J., King-Heiden, T. C., & Luoma, J. A. (2024). Seasonal differences in larval sea lamprey (*Petromyzon marinus*) sensitivity to the pesticide TFM. *Journal of Great Lakes Research*, *50*(1), 102248. https://doi.org/10.1016/j.jglr.2023.102248

Schuldt, R. J., & Goold, R. (1980). Changes in the Distribution of Native Lampreys in Lake Superior Tributaries in Response to Sea Lamprey (*Petromyzon marinus*) Control, 1953–77. *Canadian Journal of Fisheries and Aquatic Sciences*, *37*(11), 1872–1885. https://doi.org/10.1139/f80-229

Siefkes, M. J. (2017). Use of physiological knowledge to control the invasive sea lamprey (*Petromyzon marinus*) in the Laurentian Great Lakes. *Conservation Physiology*, *5*(1), cox031. https://doi.org/10.1093/conphys/cox031

Simpson, G. L. (2024). gratia: An R package for exploring generalized additive models. *Journal of Open Source Software*, *9*(104), 6962. https://doi.org/10.21105/joss.06962

Smith, B. R., & Tibbles, J. J. (1980). Sea Lamprey (*Petromyzon marinus*) in Lakes Huron, Michigan, and Superior: History of Invasion and Control, 1936–78. *Canadian Journal of Fisheries and Aquatic Sciences*, *37*(11), 1780–1801. https://doi.org/10.1139/f80-222

Sullivan, W. P., Burkett, D. P., Boogaard, M. A., Criger, L. A., Freiburger, C. E., Hubert, T. D., Leistner, K. G., Morrison, B. J., Nowicki, S. M., Robertson, S. N. P., Rowlinson, A. K., Scotland, B. J., & Sullivan, T. B. (2021). Advances in the use of lampricides to control sea lampreys in the Laurentian Great Lakes, 2000–2019. *Journal of Great Lakes Research, Supplement on Sea Lamprey International Symposium III (SLIS III)*, *47*, S216–S237. https://doi.org/10.1016/j.jglr.2021.08.009

Svendsen, J. C., Genz, J., Anderson, W. G., Stol, J. A., Watkinson, D. A., & Enders, E. C. (2014). Evidence of Circadian Rhythm, Oxygen Regulation Capacity, Metabolic Repeatability and Positive Correlations between Forced and Spontaneous Maximal Metabolic Rates in Lake Sturgeon Acipenser fulvescens. *PLOS ONE*, *9*(4), e94693. https://doi.org/10.1371/journal.pone.0094693

Biology Open • Accepted manuscript

Svendsen, M. B. S., Bushnell, P. G., & Steffensen, J. F. (2016). Design and setup of intermittent-flow respirometry system for aquatic organisms: How to set up an aquatic respirometry system. *Journal of Fish Biology*, *88*(1), 26–50. https://doi.org/10.1111/jfb.12797

Viant, M. R., Walton, J. H., & Tjeerdema, R. S. (2001). Comparative Sublethal Actions of 3-Trifluoromethyl-4-nitrophenol in Marine Molluscs as Measured by *in Vivo*31P NMR. *Pesticide Biochemistry and Physiology*, *71*(1), 40–47. https://doi.org/10.1006/pest.2001.2554

Wilkie, M. P. (2025). How the resilient ecophysiology of the sea lamprey allowed them to invade the Laurentian Great Lakes and could protect them from climate change. *Frontiers in Fish Science*, *3*. https://doi.org/10.3389/frish.2025.1591515

Wilkie, M. P., Holmes, J. A., & Youson, J. H. (2007). The lampricide 3-trifluoromethyl-4-nitrophenol (TFM) interferes with intermediary metabolism and glucose homeostasis, but not with ion balance, in larval sea lamprey (*Petromyzon marinus*). *Canadian Journal of Fisheries and Aquatic Sciences*, *64*(9), 1174–1182. https://doi.org/10.1139/f07-082

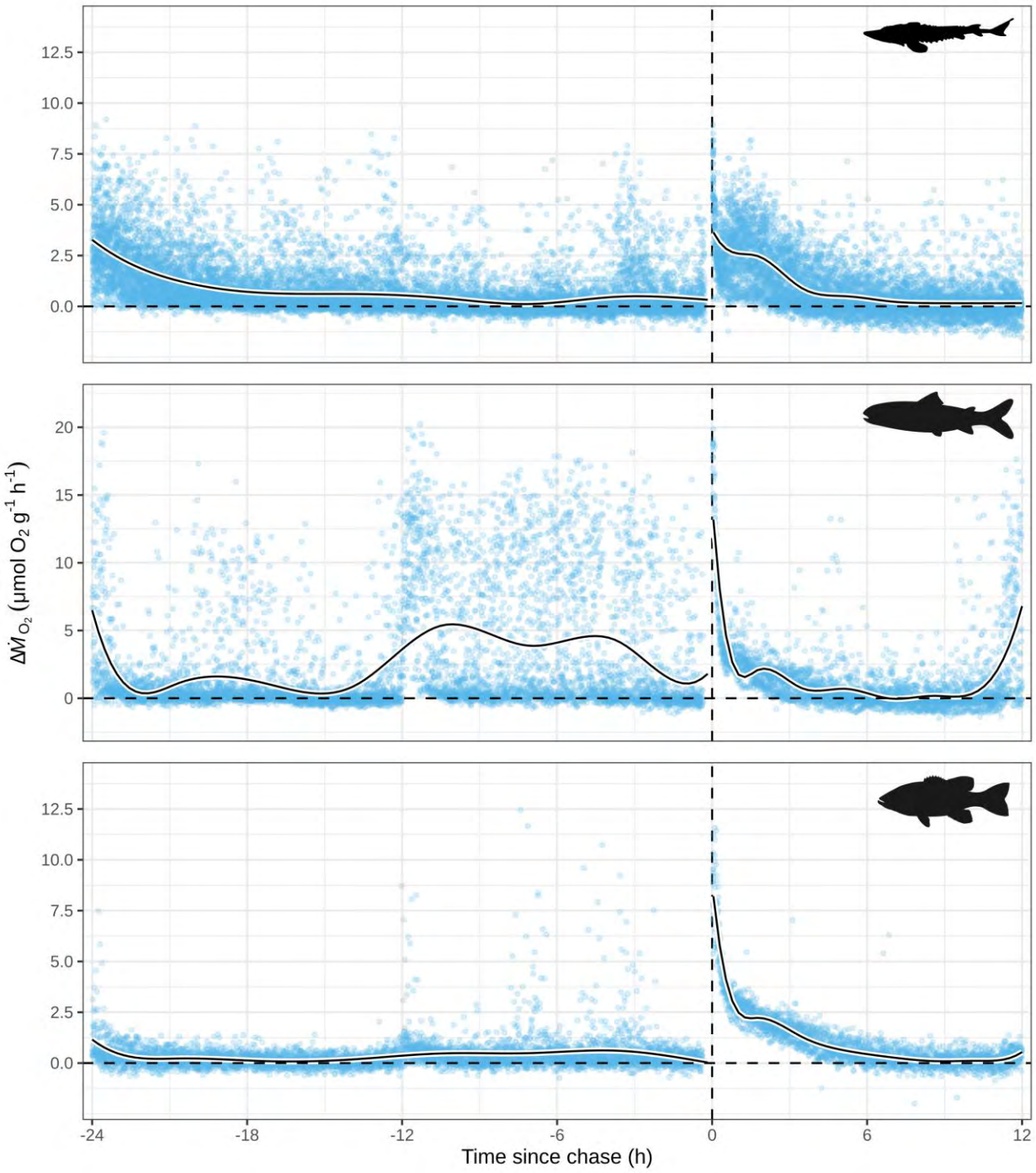
Wilkie, M. P., Hubert, T. D., Boogaard, M. A., & Birceanu, O. (2019). Control of invasive sea lampreys using the piscicides TFM and niclosamide: Toxicology, successes & future prospects. *Aquatic Toxicology*, *211*, 235–252. https://doi.org/10.1016/j.aquatox.2018.12.012

Wilkie, M. P., Johnson, N. S., & Docker, M. F. (2022). Chapter 10 - Invasive species control and management: The sea lamprey story. In N. A. Fangue, S. J. Cooke, A. P. Farrell, C. J. Brauner, & E. J. Eliason (Eds.), *Fish Physiology* (Vol. 39, pp. 489–579). Academic Press. https://doi.org/10.1016/bs.fp.2022.09.001

Zhang, Y., Claireaux, G., Takle, H., Jørgensen, S. M., & Farrell, A. P. (2018). A three-phase excess post-exercise oxygen consumption in Atlantic salmon *Salmo salar* and its response to exercise training. *Journal of Fish Biology*, *92*(5), 1385–1403. https://doi.org/10.1111/jfb.13593

Biology Open • Accepted manuscript

Figures and Tables



Biology Open • Accepted manuscript

Fig. 1. Delta oxygen consumption (Absolute $\dot{M}_{O_2}$ minus the animal's respective SMR) of lake sturgeon (n = 45), rainbow trout (n = 18), and smallmouth bass (n = 14) throughout eleven chase trials. The dashed horizontal line at $\Delta\dot{M}_{O_2}$ = 0 highlights the SMR state. The vertical dashed line indicates the point where the animals were chased to exhaustion. The solid line shows the GAM-modelled population trend over time. Chasing was conducted approximately at the same time of day every day (19:00-20:00) to minimise the effects of circadian rhythms, as those noticeable for rainbow trout. Four rainbow trout MMR values were omitted to optimize vertical plot space and ensure comparable Y axes between Figure 1 and Figure 3.

Biology Open • Accepted manuscript

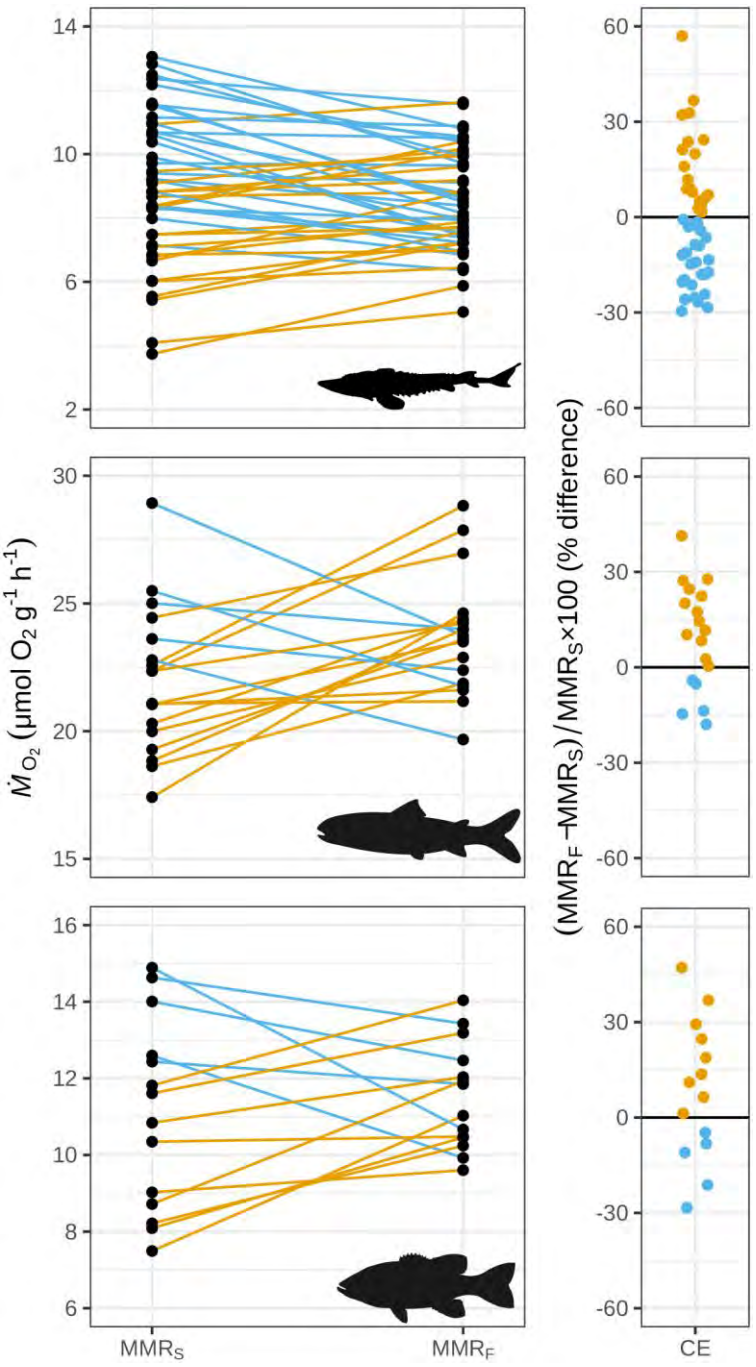


Fig. 2. Relative magnitude of the oxygen consumption rates induced by spontaneously occurring bursts of activity (MMR$_S$) and by forced chasing (MMR$_F$) for lake sturgeon (n = 45), rainbow trout (n = 18), and smallmouth bass (n = 14). Chasing efficacy (CE) is shown on the right-side panels. Positive CE values (where chasing induced higher $\dot{M}_{O_2}$ than spontaneous activity) are shown in orange, while as negative CEs are drawn in blue, with the respective relationships also drawn on the left-side panels.

Biology Open • Accepted manuscript

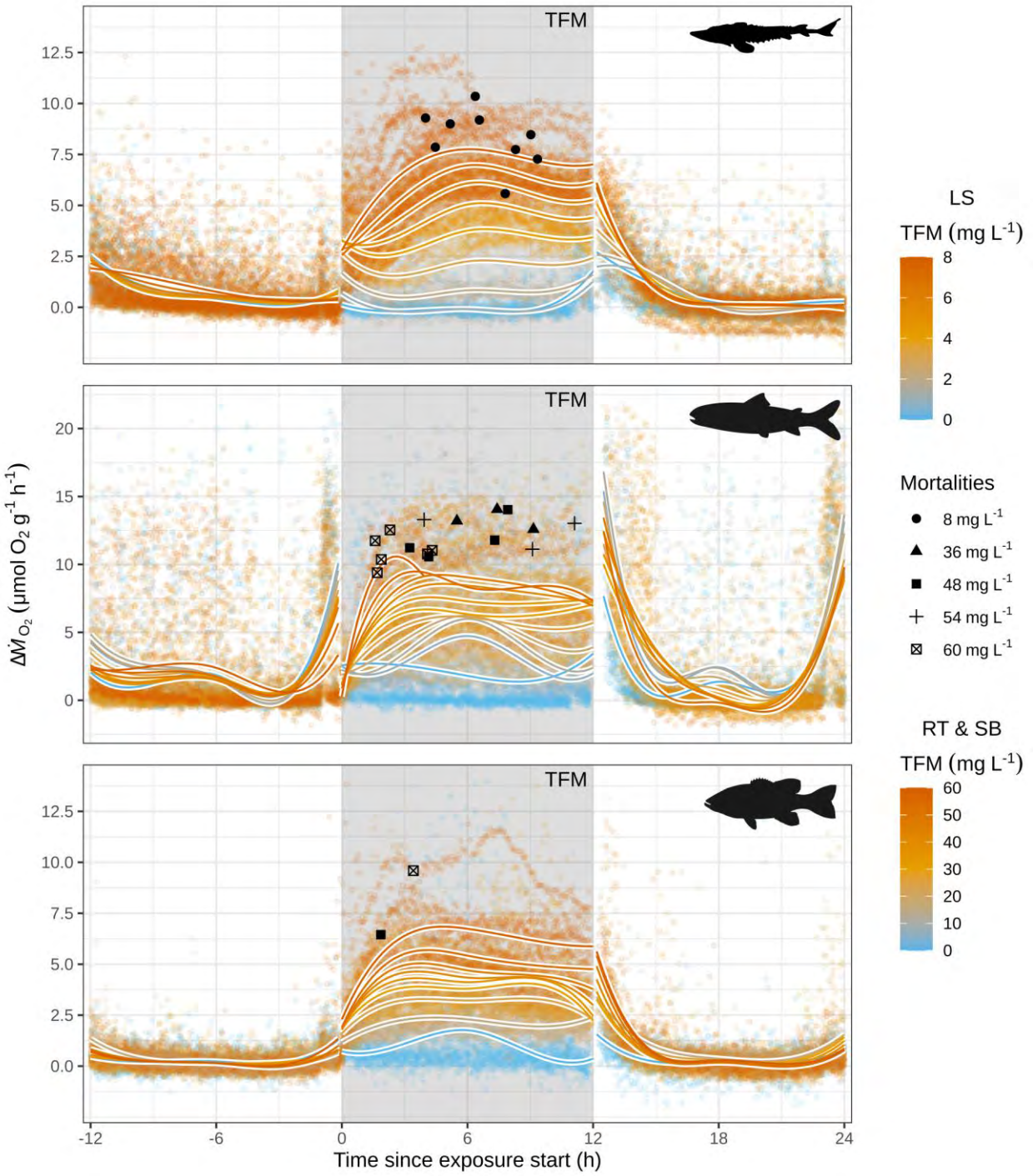


Fig. 3. Delta oxygen consumption (Absolute $\dot{M}_{O_2}$ minus the animal's respective SMR) of lake sturgeon (n = 66), rainbow trout (n = 70), and smallmouth bass (n = 60) throughout the 25 TFM toxicity trials. The coloured lines show the GAM-modelled population trends over

Biology Open • Accepted manuscript

time for the various TFM concentrations tested. TFM exposures were conducted approximately at the same time of day every day (08:00 to 20:00) to minimise the effects of circadian rhythms. Eighteen $\Delta\dot{M}_{O_2}$ values for rainbow trout and smallmouth bass were omitted to optimize vertical plot space. Note the different Y-axis ranges per panel, and the different ranges in TFM concentration for lake sturgeon (LS, top colour ramp) and rainbow trout and smallmouth bass (RT & SB, bottom colour ramp).

Biology Open • Accepted manuscript

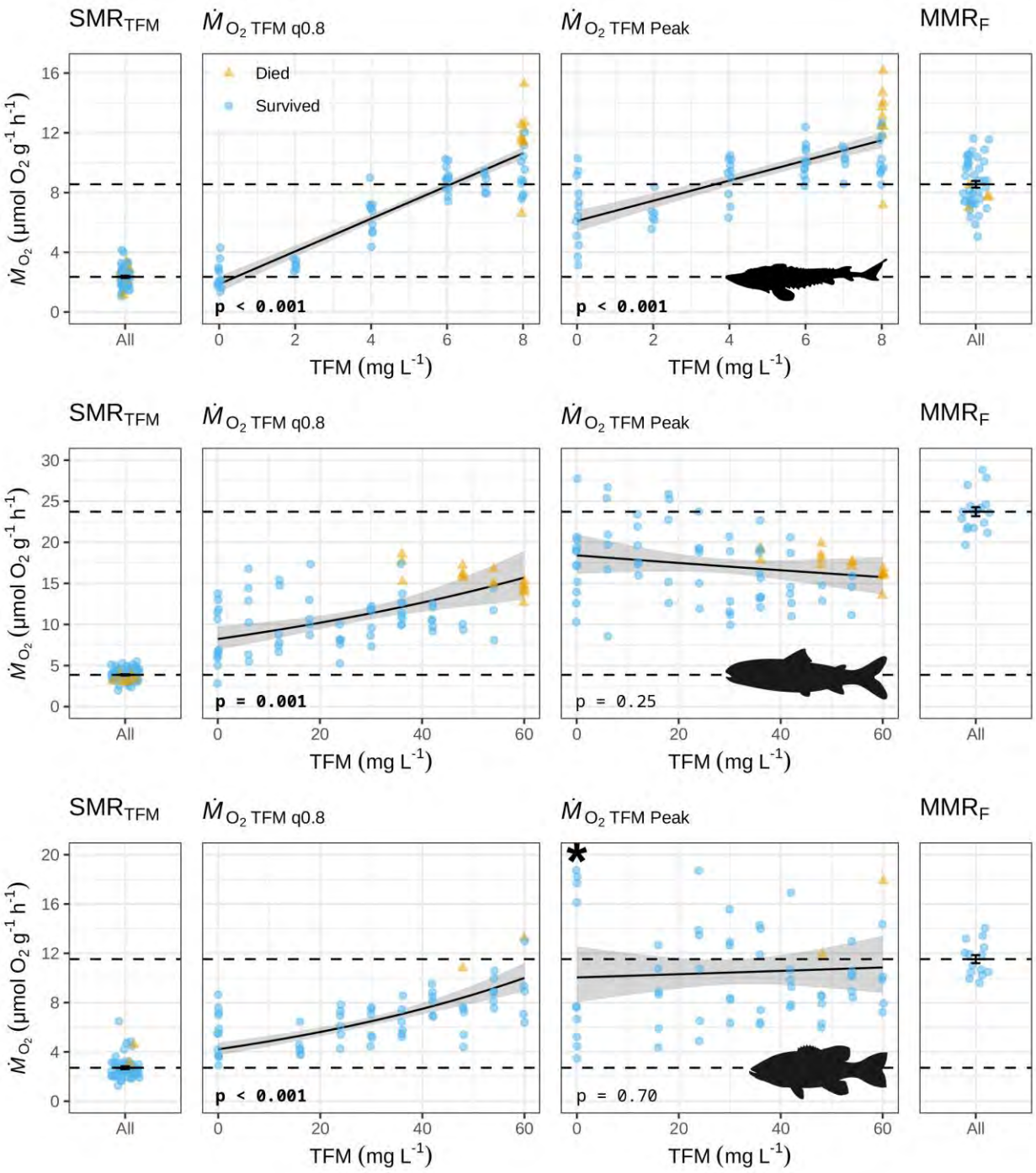


Fig. 4. Comparison of the TFM-induced changes in oxygen consumption ($\dot{M}_{O_2\text{TFM q0.8}}$ and $\dot{M}_{O_2\text{TFM Peak}}$) to the standard metabolic rate calculated prior to exposure (SMR_{TFM}) and to the maximum metabolic rate derived from chasing trials (MMR_F) for lake sturgeon,

Biology Open • Accepted manuscript

rainbow trout, and smallmouth bass. The dashed black lines illustrate how the average SMR_{TFM} and MMR_F relate to the metrics calculated during TFM exposure. The solid black lines and respective ribbons show the fitted model and respective confidence intervals of the mean. P-values indicate the significance of the relationship between each metric and TFM concentration. For more details on the models applied, please refer to Tables 1, 2, and 3. * Two high values were omitted to optimize vertical plot space.

Biology Open • Accepted manuscript

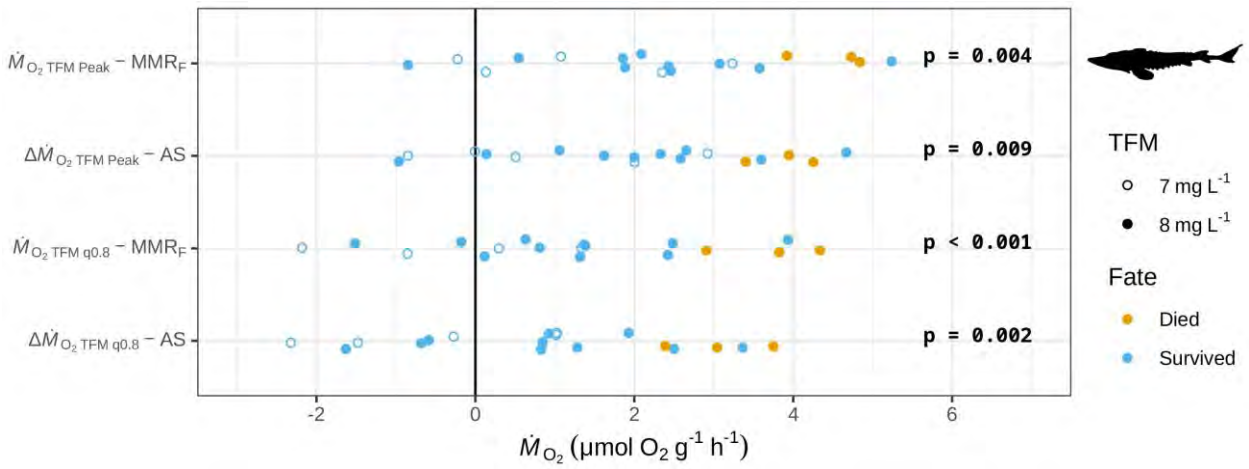


Fig. 5. Difference between TFM-induced changes in oxygen consumption ($\dot{M}_{O_2\mathbf{TFM\ q0.8}}$, $\Delta\dot{M}_{O_2\mathbf{TFM\ q0.8}}$, $\dot{M}_{O_2\mathbf{TFM\ Peak}}$, and $\Delta\dot{M}_{O_2\mathbf{TFM\ Peak}}$) and respective exhaustive chasing maximum metabolic rates (MMR_F and AS) for lake sturgeon (n = 18). The black vertical line at $\dot{M}_{O_2}$ = 0 denotes the point of no difference between the metrics calculated during TFM exposure and MMR_F/AS. P-values indicate the capacity of each of the metrics listed on the Y axis to predict the lake sturgeon's fate during TFM exposure.

Biology Open • Accepted manuscript

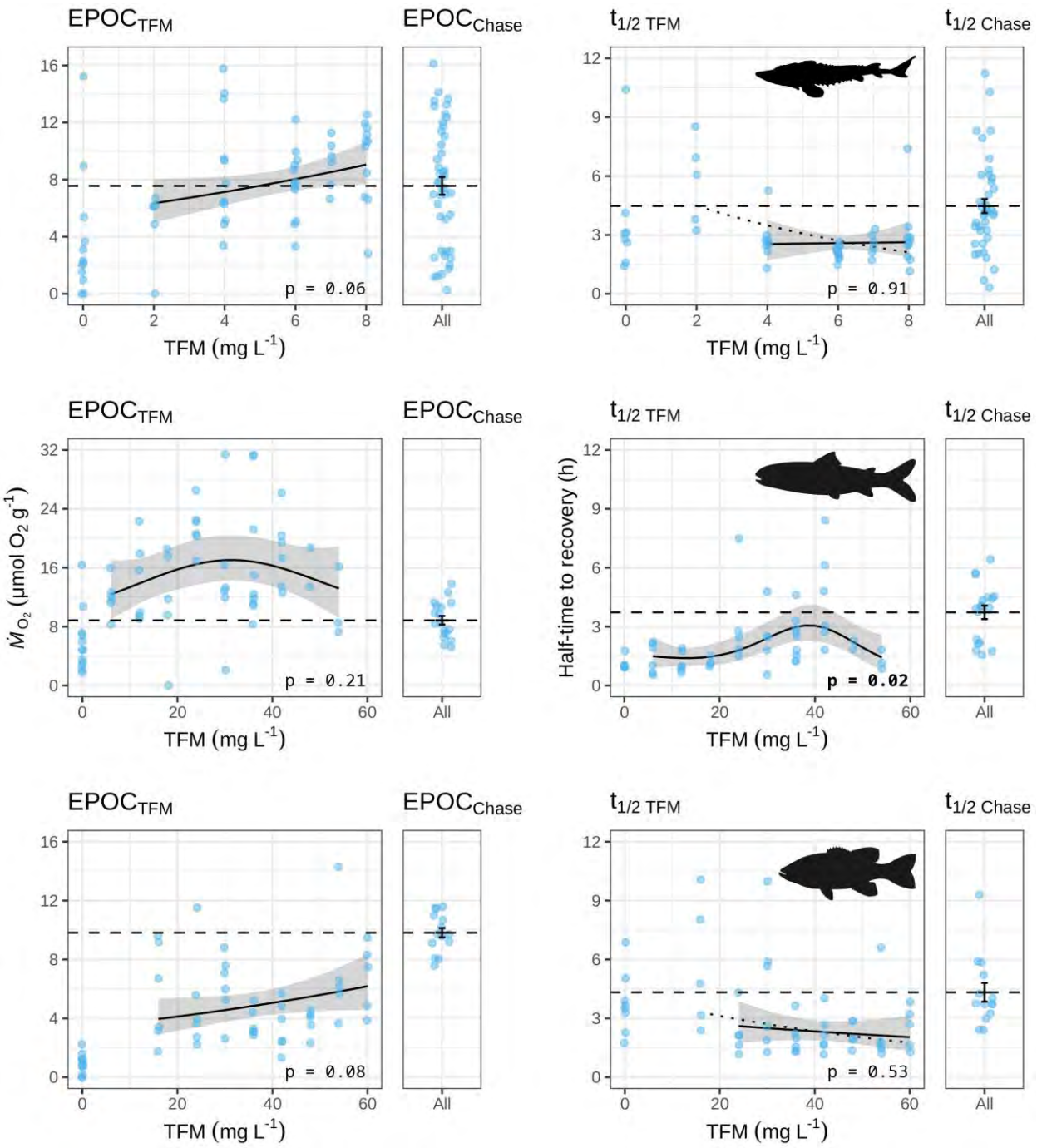


Fig. 6. Excess oxygen consumption following TFM exposure and manual chasing (left) and half-time for oxygen consumption recovery (right) for lake sturgeon (n_{Chase} = 44, n_{TFM} = 59), rainbow trout (n_{Chase} = 18, n_{TFM} = 61), and smallmouth bass (n_{Chase} = 14, n_{TFM} = 58). The dashed black lines illustrate how the chasing metrics compare to the TFM metrics. The solid black lines and respective ribbons show the fitted model and respective confidence

Biology Open • Accepted manuscript

intervals of the mean. P-values indicate the significance of the relationship between each metric and TFM concentration. For more details on the models applied, please refer to Tables 1, 2, and 3. Dotted lines shown in the $t_{1/2}$ TFM panels for lake sturgeon and smallmouth bass indicate model fits if the lowest TFM concentration data were included in the model.

Biology Open • Accepted manuscript

Table 1. Calculated metrics and their respective descriptions. Metrics obtained from the chasing trials are listed first, followed by the metrics obtained from the TFM-exposure trails.

Variable	Description
SMR_{Chase}	20th percentile of the $\dot{M}_{O_2}$ values obtained prior to chasing in the chasing trials
MMR_S	Highest spontaneous $\dot{M}_{O_2}$ measurement, prior to chasing
MMR_F	Highest $\dot{M}_{O_2}$ measurement following forced, exhaustive exercise
AS	Aerobic scope; MMR_F minus the animal's respective SMR_{Chase}
CE	Chasing efficacy (a measure of the relationship between MMR_F and MMR_S)
$EPOC_{Chase}$	Excess post-exercise oxygen consumption (chasing trials)
$t_{1/2\ Chase}$	Half-time of oxygen consumption recovery to SMR levels following chasing
SMR_{TFM}	20th percentile of the $\dot{M}_{O_2}$ values obtained prior to TFM exposure in the toxicity trials
$\dot{M}_{O_2 TFM\ q0.8}$	80th percentile of the $\dot{M}_{O_2}$ values obtained during TFM exposure
$\dot{M}_{O_2 TFM\ Peak}$	Highest $\dot{M}_{O_2}$ measurement obtained during TFM exposure
$\Delta\dot{M}_{O_2 TFM\ q0.8}$	$\dot{M}_{O_2 TFM\ q0.8}$ minus the animal's respective SMR_{TFM}
$\Delta\dot{M}_{O_2 TFM\ Peak}$	$\dot{M}_{O_2 TFM\ Peak}$ minus the animal's respective SMR_{TFM}
$EPOC_{TFM}$	Excess post-exposure oxygen consumption (toxicity trials)
$t_{1/2\ TFM}$	Half-time of oxygen consumption recovery to SMR levels following TFM exposure

Biology Open • Accepted manuscript

Table 2. Statistics summary of the models performed on the variables calculated for lake sturgeon, including the intercept (α) and slopes (β) for each response variable. Significant variables ($p < 0.05$) are highlighted in bold. Gamma models were run with log ling and Gaussian models were run with identity link. For more details on the definitions of the variables, please see Table 1.

Lake sturgeon				TFM (mg/L)			Day			Background (% of SMR)			Mass (g)			Condition factor (k)		
Variable	Model	n	α	β	F	p	β	F	p	β	F	p	β	F	p	β	F	p
SMR_{Chase}	Gamma GLM	45	0.875	-	-	-	**-0.006**	**19.9**	**< 0.001**	0.002	0.24	0.629	0.047	1.43	0.239	-0.133	0.04	0.841
MMR_F	Gamma GLM	45	2.386	-	-	-	**-0.002**	**6.11**	**0.018**	-0.003	0.84	0.365	0.048	2.63	0.113	-0.872	3.12	0.085
AS	Gamma GLM	45	2.155	-	-	-	-0.001	0.82	0.372	-0.006	2.11	0.154	0.045	1.66	0.205	-1.134	3.68	0.062
SMR_{TFM}	Gamma GLM	66	1.08	**0.026**	**7.48**	**0.008**	**-0.005**	**24.8**	**< 0.001**	**-0.007**	**9.98**	**0.002**	0.000	0.00	0.996	0.226	0.26	0.610
$\dot{M}_{O_2 TFM\ q0.8}$	Gaussian GLM	66	5.119	**1.094**	**504**	**< 0.001**	**-0.021**	**14.0**	**< 0.001**	-0.01	0.82	0.37	-0.141	0.94	0.337	-4.005	2.89	0.095
$\dot{M}_{O_2 TFM\ Peak}$	Gaussian GLM	66	10.70	**0.675**	**113**	**< 0.001**	**-0.018**	**5.77**	**0.019**	-0.014	0.91	0.343	-0.161	0.71	0.402	**-8.227**	**7.15**	**0.010**
$\Delta\dot{M}_{O_2 TFM\ q0.8}$	Gaussian GLM	66	2.261	**1.036**	**605**	**< 0.001**	-0.009	3.44	0.069	0.002	0.06	0.802	-0.146	1.35	0.25	**-4.387**	**4.64**	**0.035**
$\Delta\dot{M}_{O_2 TFM\ Peak}$	Gaussian GLM	66	7.843	**0.617**	**121**	**< 0.001**	0.006	0.75	0.390	-0.001	0.01	0.913	-0.166	0.97	0.328	**-8.609**	**10.0**	**0.002**
$EPOC_{TFM}$	Gamma GLM	44	1.488	0.059	3.67	0.063	**0.004**	**4.56**	**0.039**	0.004	0.75	0.392	-0.010	0.04	0.848	-0.045	0.00	0.958
$t_{1/2\ TFM}$ (4-8 mg L^{-1})	Gamma GLM	38	0.875	0.009	0.01	0.916	0.004	1.66	0.207	0	0	0.988	-0.093	1.63	0.211	0.441	0.14	0.713

Biology Open • Accepted manuscript

Table 3. Statistics summary of the generalized linear models performed on the variables calculated for rainbow trout, including the intercept (α) and slopes (β) for each response variable. Significant variables ($p < 0.05$) are highlighted in bold. Gamma models were run with log ling and Gaussian models were run with identity link. For more details on the definitions of the variables, please see Table 1. N.L. = non-linear.

Rainbow trout				TFM (mg/L)			Day			Background (% of SMR)			Mass (g)			Condition factor (k)		
Variable	Model	n	α	β	F	p	β	F	p	β	F	p	β	F	p	β	F	p
SMR_{Chase}	Gamma GLM	18	1.697	-	-	-	0.03	4.3	0.059	-0.001	0.27	0.609	-0.003	0.08	0.787	-0.069	0.12	0.73
MMR_F	Gamma GLM	18	3.299	-	-	-	-0.036	3.48	0.085	-0.003	0.84	0.375	-0.009	0.32	0.582	0.096	0.13	0.72
AS	Gamma GLM	18	3.07	-	-	-	-0.053	4.19	0.061	-0.003	0.59	0.457	-0.01	0.21	0.655	0.131	0.14	0.71
SMR_{TFM}	Gamma GLM	77	2.259	0.001	0.66	0.421	-0.002	1.17	0.284	**-0.014**	**4.02**	**0.049**	**-0.012**	**14.4**	**< 0.001**	**-0.513**	**10.1**	**0.002**
$\dot{M}_{O_2\,TFM\,q0.8}$	Gamma GLM	77	2.248	**0.011**	**13.3**	**0.001**	-0.006	2.75	0.102	0.017	1.5	0.225	-0.002	0.12	0.734	-0.001	0	0.998
$\dot{M}_{O_2\,TFM\,Peak}$	Gamma GLM	77	3.487	-0.003	1.36	0.248	-0.001	0.1	0.755	0.004	0.14	0.708	-0.003	0.4	0.529	-0.45	3.04	0.086
$\Delta\dot{M}_{O_2\,TFM\,q0.8}$	Gaussian GLM	77	1.94	**0.109**	**14.1**	**< 0.001**	-0.051	1.92	0.17	0.259	3.45	0.068	0.012	0.03	0.855	2.254	0.48	0.489
$\Delta\dot{M}_{O_2\,TFM\,Peak}$	Gaussian GLM	77	20.75	-0.048	1.76	0.189	-0.01	0.05	0.823	0.111	0.41	0.526	-0.011	0.02	0.891	-5.485	1.83	0.181
$EPOC_{TFM}$	Gamma GAM	48	2.079	N.L.	1.49	0.206	-0.002	0.17	0.683	-0.223	0.41	0.528	0.005	0.25	0.618	0.011	0	0.989
$t_{1/2\,TFM}$	Gamma GAM	46	-1.106	**N.L.**	**4.25**	**0.015**	0.005	0.31	0.583	-0.076	2.09	0.157	-0.004	0.08	0.773	1.909	2.32	0.136

Biology Open • Accepted manuscript

Table 4. Statistics summary of the generalized linear models performed on the variables calculated for smallmouth bass, including the intercept (α) and slopes (β) for each response variable. Significant variables ($p < 0.05$) are highlighted in bold. Gamma models were run with log ling and Gaussian models were run with identity link. For more details on the definitions of the variables, please see Table 1.

Smallmouth bass				TFM (mg/L)			Day			Background (% of SMR)			Mass (g)			Condition factor (k)		
Variable	Model	n	α	β	F	p	β	F	p	β	F	p	β	F	p	β	F	p
SMR_{Chase}	Gamma GLM	14	1.078	-	-	-	0.05	4.80	0.056	0.001	0.67	0.435	-0.038	4.76	0.057	-0.065	0.06	0.812
MMR_F	Gamma GLM	14	1.869	-	-	-	0.017	0.88	0.373	0.001	0.12	0.736	**0.05**	**12.3**	**0.007**	0.251	1.34	0.277
AS	Gamma GLM	14	1.412	-	-	-	0.009	0.16	0.700	0	0.02	0.897	**0.075**	**20.1**	**0.002**	0.338	1.79	0.213
SMR_{TFM}	Gamma GLM	59	2.143	0	0.04	0.850	0	0.05	0.818	-0.002	0.51	0.480	**-0.06**	**32.3**	**< 0.001**	**-0.641**	**56.5**	**< 0.001**
$\dot{M}_{O_2 TFM\ q0.8}$	Gamma GLM	59	2.346	**0.014**	**63.6**	**< 0.001**	0.002	0.86	0.357	-0.004	1.42	0.238	**-0.054**	**13.2**	**0.001**	**-0.470**	**15.3**	**< 0.001**
$\dot{M}_{O_2 TFM\ Peak}$	Gamma GLM	59	3.365	0.001	0.15	0.700	0.006	1.46	0.232	-0.007	1.10	0.299	**-0.082**	**8.3**	**0.006**	**-0.503**	**5.36**	**0.024**
$\Delta\dot{M}_{O_2 TFM\ q0.8}$	Gaussian GLM	59	3.482	**0.101**	**83.3**	**< 0.001**	0.013	0.79	0.379	-0.030	2.17	0.147	**-0.209**	**5.21**	**0.027**	-0.968	1.69	0.199
$\Delta\dot{M}_{O_2 TFM\ Peak}$	Gaussian GLM	59	16.70	0.003	0.01	0.931	0.054	1.04	0.313	-0.045	0.39	0.534	-0.639	3.97	0.052	-4.597	3.10	0.084
$EPOC_{TFM}$	Gamma GLM	45	1.617	0.01	3.19	0.082	**0.024**	**10.8**	**0.002**	-0.005	0.36	0.551	-0.024	0.39	0.538	-0.581	2.19	0.147
$t_{1/2\ TFM}$ (24-60 mg L^{-1})	Gamma GLM	38	0.616	-0.007	0.41	0.528	0.006	0.36	0.553	0.005	0.18	0.673	0.016	0.08	0.774	0.151	0.1	0.749

Biology Open • Accepted manuscript

Table S1. Summary information of all the experimental trials conducted in this project.

Date	Exp. type	Species	Temperature	Alkalinity	pH	$TFM_{nominal}$	$TFM_{observed}$
2025-11-09	MMR	RT, SB	14.8 (14.6-15.0)	n.a.	n.a.	-	-
2025-11-11	MMR	RT, SB	15.1 (15.0-15.1)	75	n.a.	-	-
2025-11-13	MMR	RT, SB	15.1 (15.0-15.2)	81	n.a.	-	-
2025-11-16	MMR	LS	15.0 (14.9-15.2)	92	n.a.	-	-
2025-11-18	MMR	LS	15.0 (14.9-15.0)	120	n.a.	-	-
2025-11-20	MMR	LS	15.0 (14.9-15.0)	177	n.a.	-	-
2025-11-23	TFM	LS	15.0 (14.9-15.1)	175	8.58 (8.56-8.60)	0	0
2025-11-25	MMR	LS	15.0 (14.9-15.0)	177	8.56 (8.49-8.59)	-	-
2025-11-27	TFM	LS	15.0 (14.7-15.1)	180	8.54 (8.53-8.55)	8	8.1 (7.9-8.3)
2025-11-30	TFM	LS	15.0 (14.8-15.1)	177	8.61 (8.58-8.62)	4	4.1 (4.0-4.2)

Table S1. Continued.

Date	Exp. type	Species	Temperature	Alkalinity	pH	$TFM_{nominal}$	$TFM_{observed}$
2025-12-02	TFM	RT	14.9 (14.7-15.0)	186	8.53 (8.45-8.57)	12	12.0 (11.9-12.3)
2025-12-04	TFM	RT	14.9 (14.7-15.1)	196	8.46 (8.41-8.50)	18	18.1 (17.6-18.3)
2025-12-07	TFM	RT, SB	14.9 (14.8-15.0)	185	8.66 (8.63-8.67)	0	0
2025-12-09	TFM	LS, RT	14.9 (14.5-15.0)	180	8.56 (8.52-8.59)	6	6.0 (5.8-6.3)
2025-12-14	TFM	RT, SB	15.0 (14.9-15.0)	184	8.63 (8.59-8.66)	0	0
2025-12-16	TFM	RT, SB	15.0 (14.8-15.0)	n.a.	8.55 (8.51-8.58)	36	36.1 (35.6-36.6)
2025-12-18	TFM	RT, SB	15.0 (14.9-15.1)	185	8.54 (8.49-8.57)	42	42.0 (40.7-43.0)
2026-01-04	TFM	RT, SB	15.1 (14.9-15.3)	182	8.57 (8.55-8.59)	30	30.1 (28.9-32.1)
2026-01-06	TFM	RT, SB	15.0 (14.8-15.2)	182	8.54 (8.48-8.57)	48	47.9 (46.9-48.6)
2026-01-08	TFM	RT, SB	15.0 (14.9-15.1)	176	8.54 (8.50-8.55)	54	54.1 (53.2-55.9)
2026-01-11	TFM	RT, SB	15.0 (14.9-15.1)	177	8.37 (8.32-8.40)*	24	24.2 (23.9-24.7)
2026-01-13	TFM	RT	15.0 (14.9-15.0)	n.a.	8.36 (8.31-8.40)*	36	36.0 (34.9-37.0)
2026-01-15	TFM	RT, SB	15.0 (14.9-15.1)	185	8.40 (8.35-8.44)*	60	59.3 (57.9-60.6)

* The pH meter readings were lost for these exposure periods. The data from the sensor spots is available, but the high concentration of TFM in these trials caused considerable interference in the sensor spot readings during the exposure period. Therefore, the best approximation available, listed in these cells, is the summary of the sensor-spot readings in the pre-exposure period. These are likely slightly below the values that would have been recorded by the pH meter (by approximately 0.2 based on the remaining data) as the sensor spots gradually lost sensitivity with consecutive exposures to high concentrations of TFM.

Table S1. Continued.

Date	Exp. type	Species	Temperature	Alkalinity	pH	$TFM_{nominal}$	$TFM_{observed}$
2026-01-18	TFM	LS	15.0 (14.9-15.1)	185	8.62 (8.58-8.64)	6	5.9 (5.7-6.0)
2026-01-20	TFM	SB	15.0 (14.9-15.2)	191	8.55 (8.50-8.57)	16	15.99
2026-01-25	MMR	LS	15.1 (14.6-15.2)	177	8.55 (8.45-8.58)	-	-
2026-01-27	MMR	LS	15.0 (15.0-15.1)	177	8.53 (8.51-8.55)	-	-
2026-01-29	MMR	LS	15.0 (14.9-15.1)	182	8.52 (8.48-8.57)	-	-
2026-02-01	MMR	LS	15.0 (14.9-15.0)	176	8.64 (8.62-8.65)	-	-
2026-02-03	TFM	LS	14.9 (14.8-15.0)	184	8.54 (8.51-8.55)	0	0
2026-02-05	TFM	LS	15.0 (14.7-15.1)	186	8.51 (8.47-8.53)	4	3.99
2026-02-15	TFM	LS	15.0 (14.9-15.1)	180	8.61 (8.57-8.65)	2	1.98
2026-02-17	TFM	LS	15.0 (14.9-15.1)	181	8.42 (8.38-8.44)	7	6.97
2026-02-19	TFM	LS	15.0 (14.9-15.1)	179	8.57 (8.54-8.59)	8	7.99
2026-02-22	TFM	LS	15.1 (15.0-15.3)	n.a.	8.59 (8.54-8.62)	8	8.09
2026-02-24	TFM	LS	15.1 (15.0-15.1)	181	8.52 (8.49-8.53)	8	8.05

Biology Open • Supplementary information

Biology Open • Supplementary information

Supplementary Material 2: Photos/schematics of the experimental system

The experimental system design followed that described in Supplementary Material 1 of D'Souza et al (2025, doi: 10.1016/j.jglr.2025.102536). Summarily, the system consists of two independent recirculation systems, one running clean water and the other dosed with TFM, as shown in the schematic presented in the next page. Some photos are also provided.

Schematic of the wet table:

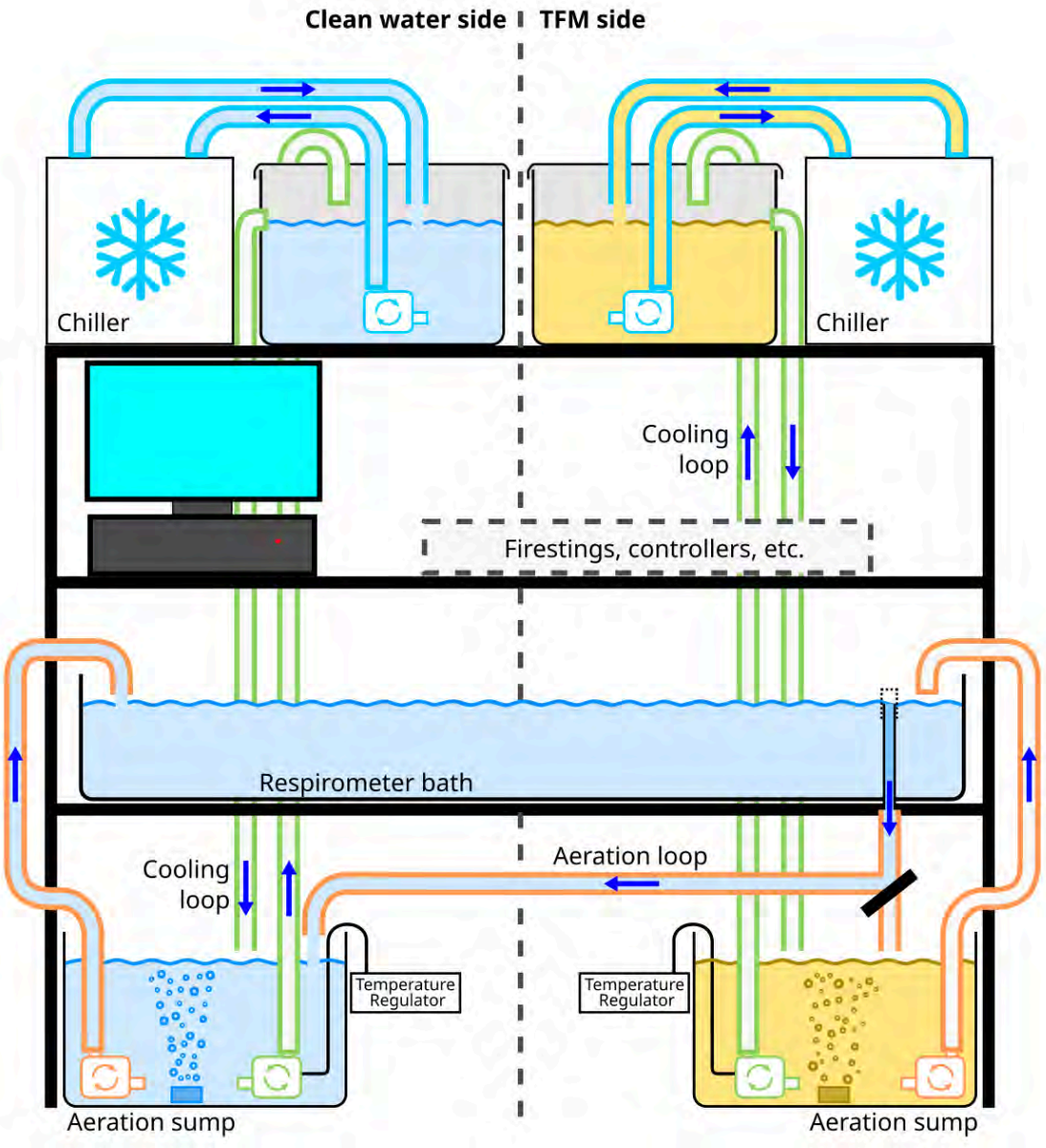
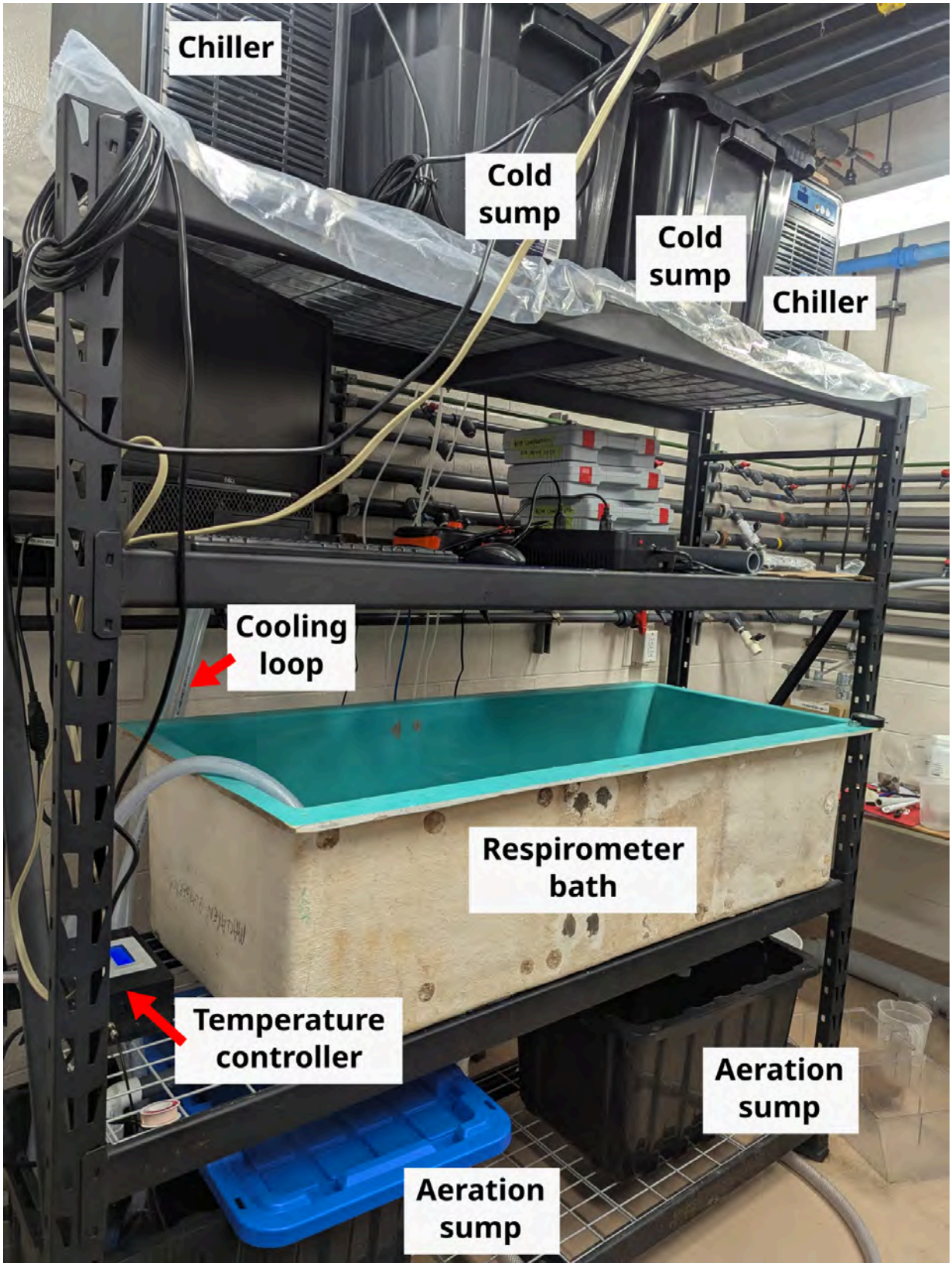
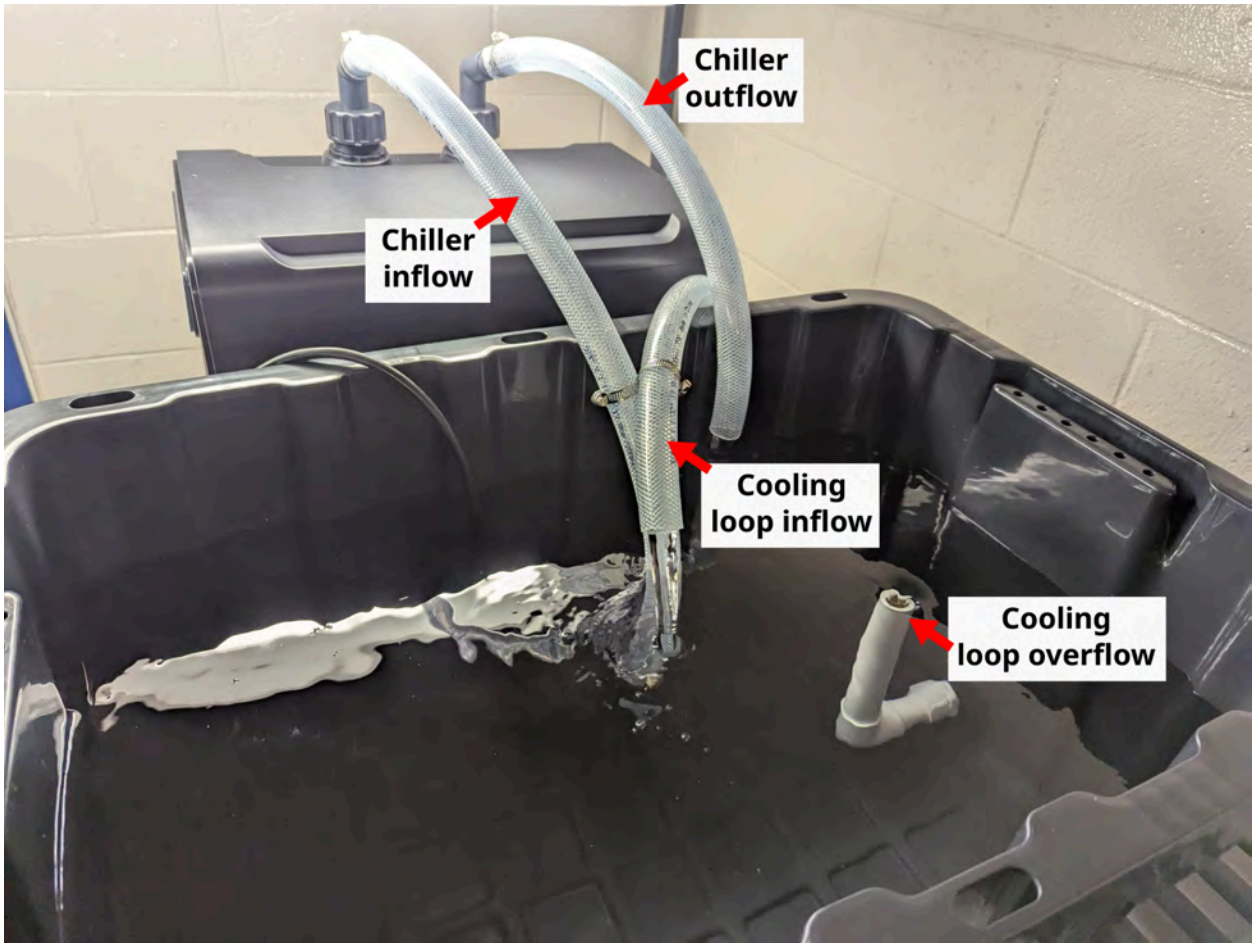


Photo of the wet table during assembly:



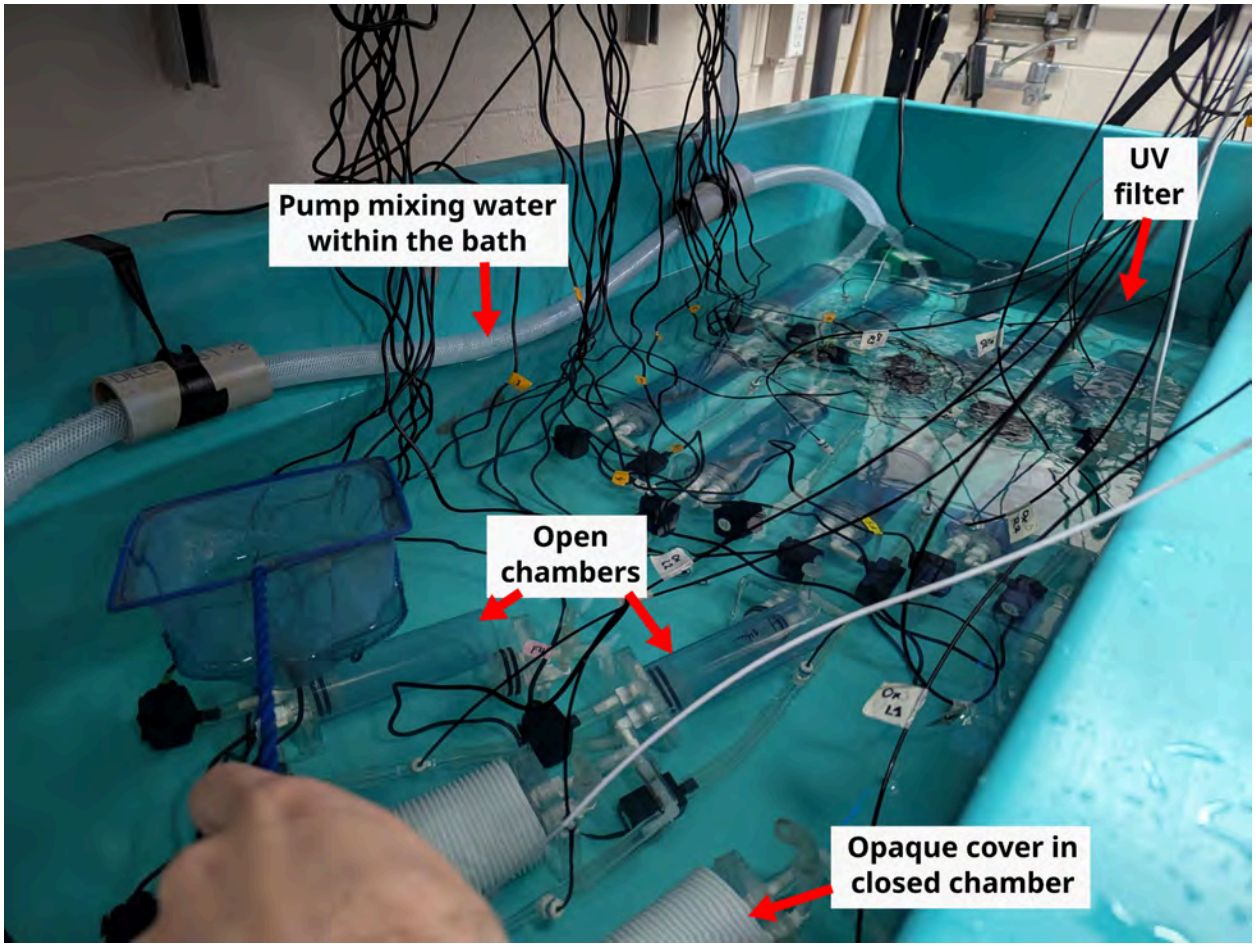
Biology Open • Supplementary information

Close up of the clean water cooling sump:



Biology Open • Supplementary information

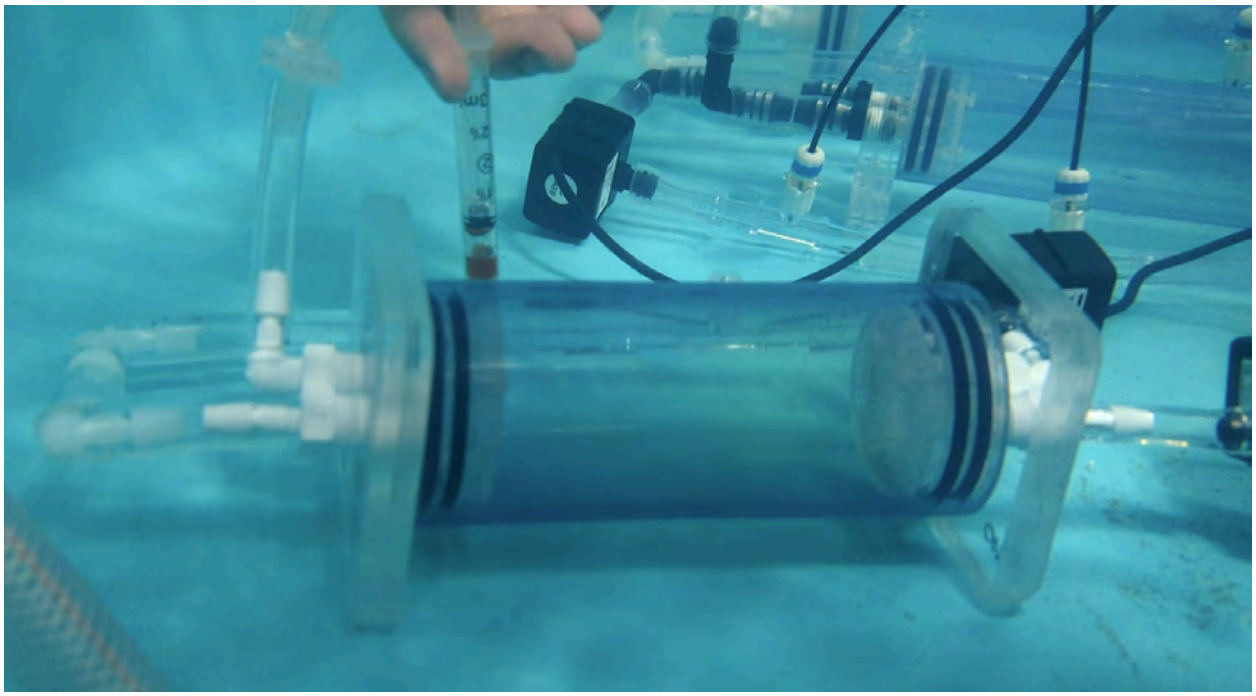
Respirometer bath with clean water (before trial start):

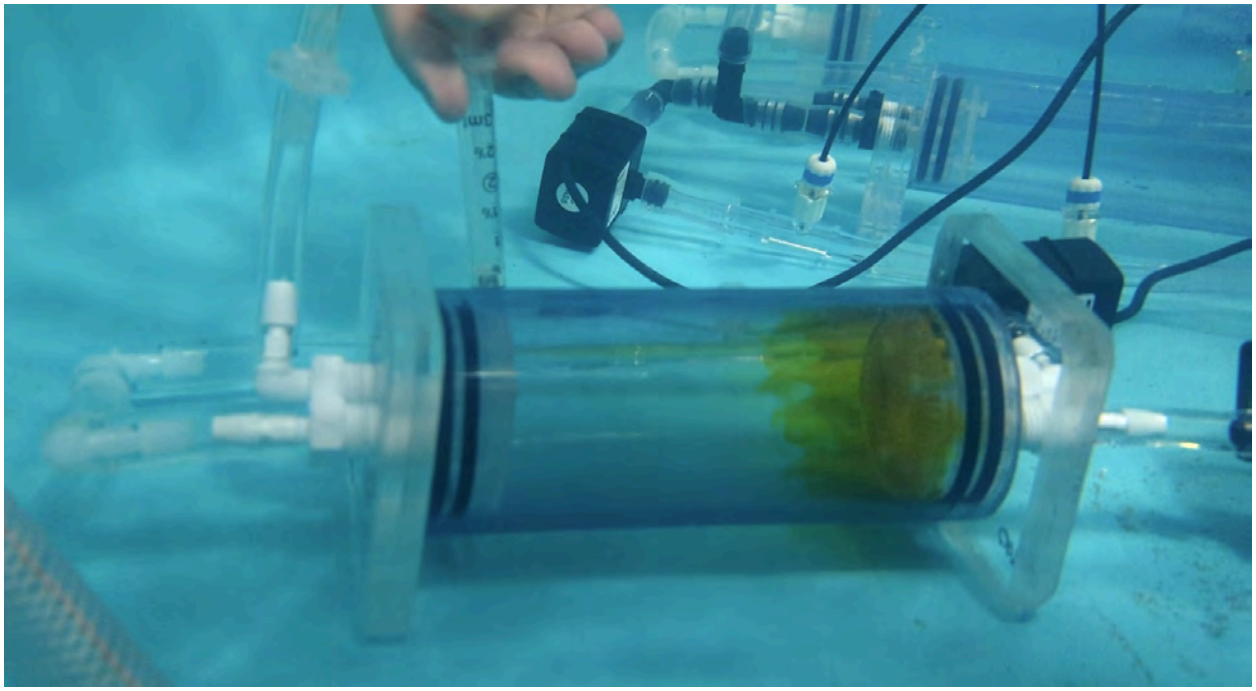


UV filters were turned off during the TFM exposure periods.

Dye test sequence on one chamber:

20s video available in the paper's Zenodo repository: www.doi.org/10.5281/zenodo.21877281





Biology Open • Supplementary information

